

# An adaptive error-controlled hybrid fast solver for regularized vortex methods

Samer Salloum, Issam Lakkis \*

Department of Mechanical Engineering, American University of Beirut, Beirut, Lebanon

## ARTICLE INFO

### Article history:

Received 29 January 2021

Received in revised form 5 July 2022

Accepted 22 July 2022

Available online 28 July 2022

### Keywords:

Vortex methods

Fast multipoles

Adaptive fast solver

## ABSTRACT

In this paper, an adaptive error-controlled hybrid fast solver that combines both  $\mathcal{O}(N)$  and  $\mathcal{O}(N \log N)$  schemes is proposed. For a given accuracy, the adaptive solver is used in the context of regularized vortex methods to optimize the speed of the velocity and vortex stretching calculation. This is accomplished by introducing criteria for cell division in building of the tree, conversion of multipole to local expansion in the downward pass, stopping of the downward pass and choosing between direct and fast summation to compute the vector fields. These criteria are based on key parameters  $(p, n_F, n_T, d_\sigma)$  which take into account the elements distribution, choice of the regularization function, and the computer architecture. The proposed solver automatically adapts to the evolving flow-field by periodically updating the optimal values of these parameters to maximize the speed, while meeting the accuracy constraints, by balancing far and near-field calculations. Performance of the proposed scheme is investigated in terms of the dependence of cost and accuracy on the various controlling parameters. The evolution of the optimal values of these parameters along with the associated computational savings are presented for the case of collision of two vortex rings over a reasonable time span.

© 2022 Elsevier Inc. All rights reserved.

## 1. Introduction

Vortex methods are grid-free methods used to numerically solve the Navier-Stokes equations by following the evolution of vorticity-carrying fluid elements. In these methods, computing the velocity and vortex stretching vectors at a given position is done by summing the contributions of all vortex elements in the domain. Thus direct evaluation of these vectors at the positions of  $N$  elements requires  $\mathcal{O}(N^2)$  computational operations. The prohibitive cost of large  $N$  simulations [13] led to the development of approximate solutions for  $N$  body problems.

Two variants of hierarchical  $N$ -body methods have been developed, Treecodes and Fast Multipoles Methods (FMMs). To reduce the complexity to  $\mathcal{O}(N \log N)$ , the Treecode algorithm [1] clusters source elements into progressively larger bins and then approximate their contributions at the targets locations using multipole expansions. In addition to clustering source elements, the FMM [6] reduces the complexity to  $\mathcal{O}(N)$  by clustering the nearby target elements as well. The  $\mathcal{O}(N)$  FMM scheme then employs local expansions to approximate the contributions of the source elements at the targets locations [13]. These two methods followed distinct paths of evolution and differ from each other in the following ways [4,5,13,12,11]. (i) For the far field, Treecodes perform cell-element interactions resulting in  $\mathcal{O}(N \log N)$  complexity, while FMMs perform cell-cell interactions resulting in  $\mathcal{O}(N)$  complexity. (ii) Treecodes conventionally use hard coded Cartesian expansions with a

\* Corresponding author.

E-mail address: il01@aub.edu.lb (I. Lakkis).

fixed order of expansion (e.g.  $p = 3$ ), while FMMs employ spherical harmonic expansions of an arbitrary order. (iii) To build the interaction list, Treecodes utilize the ratio,  $\theta$  (also known as the multipole acceptance criterion (MAC)), of the cells size to the distance between cells to decide whether or not to further subdivide a cell. In contrast, FMMs use relationships between the parent, child, and neighbor to build the interaction list [13]. (iv) In Treecodes, the accuracy is controlled by the value of  $\theta$  (MAC), while in FMMs, the order of expansion  $p$  is the controlling parameter. (v) In Treecodes, the expansions are around the centers of mass of the cells, while in FMMs, they are around at the geometrical centers [5]. (vi) While Treecodes are adaptive in nature and permit interaction of cells at different levels, FMMs usually employ a non-adaptive structure of hierarchical grids where only interactions of cells at the same level are considered [4].

There have been several efforts to take the best of these two methods and to develop optimal algorithms [12]. Cheng et al. [3] extended the work of Greengard [6] to allow for adaptive interaction of cells at different levels in the tree and introduced a mechanism for choosing between cell-cell, cell-element, and element-element interaction. To decide on the type of interaction to be executed, Cheng associated with each cell four different lists of cells, which are assembled as the oct-tree adapts to the evolving particles distribution. Element-element interactions are carried out if the number of elements in a given source cell is less than  $p^2$ . Although this criterion captures the expected dependence on  $p$ , it is based on a non-optimal value of the critical number of elements in a cell below which direct interaction should be carried out. Moreover, the Oct-tree depth is controlled by the maximum number of elements per cell,  $N_c$ , which is specified by the user. Choosing a large value of  $N_c$  results in a shallow tree structure where direct interaction with the neighboring cells will likely yield significant increase in the computational cost. On the other hand, if  $N_c$  is too small, the scheme will manage to balance cell-cell and element-element interactions, but it will incur the additional cost of evaluating multipole and local expansions coefficients at the centers of cells of high level, although these coefficients are not needed to evaluate the approximate field. In addition, Cheng et al. did not discuss the limitations arising from the use of a regularization function and the dependence on the implementation in different hardware architectures.

Walter Dehnen [4,5] extended the Treecode method of Barnes and hut [1] for fast evaluation of gravitational forces by including mutual cell-cell interactions, dual tree traversal, and a mass-dependent error-controlled multipole acceptance criteria based on the upper-bound expression of the Newtonian forces. To evaluate cell-cell interactions, Dehnen uses the expressions presented in [8], which were derived from Taylor series expansion in Cartesian coordinates. Dehnen [4] also introduced five different numbers, namely,  $N_{cb}^{pre}$ ,  $N_{cb}^{post}$ ,  $N_{cc}^{pre}$ ,  $N_{cc}^{post}$ ,  $N_{cs}$ . While  $N_{cs}$  is used to decide whether or not to divide a cell to carry out Cell-self interactions, the remaining four numbers are used to determine the type of interaction to perform between distinct cells, choosing between cell-cell, cell-body, and body-body interaction. The author determined, through experiments, the optimal values of these five numbers that would result in the most efficient code for a given accuracy and given hardware architecture, but he did not present an automated algorithm for adapting to different hardware architecture or desired level of accuracy. Cell-cell and cell-body interactions are approximated using a Taylor series expansion with a fixed order  $p = 3$ , so that the MAC number is the only accuracy-control parameter. Upon comparing the performance of his code with that of the 3D adaptive FMM code by Cheng et al. [3], Dehnen [4] showed that, in the low accuracy regime, his code is more than ten times faster and twice as accurate, which makes it favorable for applications where low accuracy is traded for speed (cost), such as collisionless stellar dynamics. This is in agreement with earlier findings that report on the performance of traditional FMM in these low accuracy applications [2]. For high accuracy (e.g.  $O(10^{-6})$ ), however, Cheng et al. FMM outperforms Dehnen's code in terms of speed. According to Dehnen, this behaviour is attributed to the fact that, while his code increases accuracy by decreasing the MAC number  $\theta$  ( $p$  is always constant), FMM increases the accuracy by increasing  $p$ . He concluded by pointing out that a code for which  $p$  and  $\theta$  can be adapted simultaneously would be superior to both codes.

Yokota and Barba [14] extended Dehnen's method by implementing a hybrid Treecode-FMM algorithm that offers control over both the order of expansion and MAC. Depending on the required accuracy, their code switches between Cartesian expansions (lower accuracy) and spherical harmonic expansions (higher accuracy). In addition, their algorithm is capable of auto-tuning the kernels on heterogeneous architectures. The key for auto-tuning is to time all the kernels. The algorithm uses this information to optimally choose the type of interaction to execute in order to evaluate the mutual contribution between non-neighboring cells during the dual tree traversal. However, Yokota did not introduce any mechanism to minimize the computational cost of calculating the interaction between neighboring cells (including cell-self interactions). In fact, the interaction between neighboring cells is performed through direct summation if both cells are at the terminal tree level, that is the number of elements in these terminal cells is less than  $N_c$  where the value of  $N_c$  is introduced by the user. A non-optimal value of  $N_c$  limits the efficiency of the method. Choosing  $N_c$  to be too small results in the additional cost of building the tree and computing the expansion coefficients for a deeper tree. On the other hand, choosing  $N_c$  to be too large increases the cost due to the associated element-element interactions. Moreover, Yokota and Baraba did not address the use of a regularization core function and the additional cost incurred to control the regularization error and to maintain a certain level of accuracy.

Yokota and Barba tested their code by comparing its performance to a pure FMM code while fixing the maximum number of elements per leaf,  $N_c$ . For a well chosen value of  $N_c$  ( $N_c = 200$ ), the results show that the hybrid method always favors cell-cell interactions and, as such, does not offer a clear advantage over the pure FMM. For  $N_c = 50$ , while the FMM suffers from load imbalance between the far and near-field calculations, the hybrid method performs much better because it can choose to carry out element-element interactions even if the cell is not a leaf. Yokota and Barba concluded that their hybrid method yields better results due to the automatic fine tuning of the balance between cell-cell, cell-element, and element-

element interactions throughout the adaptive tree. The hybrid method removes the wavy behavior of the dependence of the CPU time on  $N$  in pure FMMs [14] and achieves optimum performance for all  $N$ . Yokota and Barba also investigated the effect of changing both the multiple acceptance criteria  $\theta$  and the order of expansion  $p$  to achieve maximum performance for a given accuracy on both CPUs and GPUs [12]. They showed that it is inefficient to increase  $\theta$  to a value that is larger than that of standard FMM codes ( $\theta = 1/2$ ), for both CPU and GPU implementations. Therefore, when designing a hybrid Treecode-FMM algorithm there is no need for a variable MAC since its optimal value is always  $1/2$ . In this case, the hybrid scheme, developed by Yokota and Barba, yields a similar interaction list as the algorithm proposed by Cheng et al. in [3]. The implementation of the hybrid scheme is, however, much simpler and the algorithm is more general and flexible.

As for ensuring a desired accuracy, we emphasize the work of Winckelmans [10], who used the Treecode algorithm to simulate viscous and inviscid three dimensional flows using the regularized vortex particle method. In his algorithm, Winckelmans controlled the error incurred in approximating the velocity vector by a truncated Taylor series expansions of a fixed order  $p = 2$ . For a desired accuracy, Winckelmans used the upper bound expression of the error of the velocity field to evaluate a critical value of the distance between the source cell and target element,  $d_{crit}$ , above which the multipole expansion approximation is used. We note that, using an upper bound of the error in the velocity field calculation incurs an additional computational cost, which may be excessively large when compared to using an accurate measure of the error. Moreover, and in order to minimize the regularization error, Winckelmans employed a cutoff distance equal to  $5\sigma$  below which only direct summation is used, where  $\sigma$  is the smoothing radius of the regularization function.

In this paper, we build on the work of Greengard [6] to introduce an error-controlled hybrid method that automatically adapts to the time evolution of the particles distribution to minimize the cost of computing the velocity vector and/or rate of strain tensor fields, while meeting a pre-specified accuracy constraint. The proposed algorithm accomplishes this goal by automatically carrying out periodic load-balancing of the cell-cell, cell-element, and element-element computations as the flow-field evolves in time. To realize optimal load, we introduce various criteria to decide on the order of expansion, tree depth, and whether to compute multipole expansions, local expansions or further divide a cell in the downward and upward tree passes. The criteria are based on key control parameters, whose optimal values not only depend on the evolving elements distribution, but also on the choice of the core function, on the computer architecture, and specific implementation of the executable code.

This paper is organized as follows. The vorticity transport equation and representation of the velocity and vorticity fields in terms of singular and regularized elements are presented in section 2. In section 3, we introduce the adaptive fast multipole algorithm along with the key parameters employed in tree division and decision making for load balancing. Section 4 shows how the key parameters are evaluated in terms of the order of expansion using cost analysis and time measurements specific to computer architecture, code implementation, choice of regularization function, and identity of the vector quantity being evaluated. Decomposition of the error, derivation of upper bounds of the constituting components, and the proposed error-controlled algorithm are the subject of section 5. In the results and discussion section (section 6), we report on the computational cost, investigate the dependence of the error on key parameters, and assess the performance of the algorithm in simulating the collision of two vortex rings. In section 7, we conclude and discuss future directions.

## 2. Vortex methods and the vorticity transport equation

For a three dimensional incompressible flow,  $\nabla \cdot \vec{u} = 0$ , the transport of vorticity,  $\vec{\omega}$ , is governed by the Helmholtz's vorticity equation:

$$\frac{d\vec{\omega}}{dt} = (\vec{\omega} \cdot \nabla) \vec{u} + \frac{\nabla \rho \times \nabla P}{\rho} + \nabla^2 \vec{\omega} \quad (1)$$

where the velocity vector field,  $\vec{u}$ , for an incompressible flow in an unbounded domain, is calculated as the curl of a vector potential field,  $\vec{\psi}$

$$\vec{u} = \nabla \times \vec{\psi} \quad (2)$$

Noting that the vorticity vector field is the curl of the velocity vector field, we obtain:

$$\vec{\omega} = \nabla \times \vec{u} = \nabla \times (\nabla \times \vec{\psi}) = -\nabla^2 \vec{\psi} + \nabla(\nabla \cdot \vec{\psi}) \quad (3)$$

For  $\nabla \cdot \vec{\psi} = 0$ , the vector potential field,  $\vec{\psi}$ , is governed by the Poisson equation

$$\nabla^2 \vec{\psi} = -\vec{\omega} \quad (4)$$

In an unbounded domain, the Green's function of the Poisson equation (4) is  $G(\vec{x}) = -\frac{1}{4\pi \|\vec{x}\|}$ , and the vector potential field,  $\vec{\psi}$ , is calculated as

$$\vec{\psi}(\vec{x}, t) = \frac{1}{4\pi} \int \frac{\vec{\omega}(\vec{y}, t)}{\|\vec{x} - \vec{y}\|} d^3 y \quad (5)$$

For a singular (point vortices) representation of the vorticity field,

$$\vec{\omega}_\delta(\vec{x}, t) = \sum_i \vec{\alpha}_i \delta(\vec{x} - \vec{x}_i), \quad (6)$$

the potential and velocity vector fields are approximated as

$$\vec{\psi}_\delta(\vec{x}, t) = \frac{1}{4\pi} \sum_i \frac{\vec{\alpha}_i(t)}{\|\vec{x}(t) - \vec{x}_i(t)\|}, \text{ and} \quad (7)$$

$$\vec{u}_\delta(\vec{x}, t) = \nabla \times \vec{\psi}_\delta = -\frac{1}{4\pi} \sum_i \frac{(\vec{x}(t) - \vec{x}_i(t))}{\|\vec{x}(t) - \vec{x}_i(t)\|^3} \times \vec{\alpha}_i(t) \quad (8)$$

The vortex stretching vector  $(\vec{\omega} \cdot \nabla) \vec{u}$  in equation (1) is equal to the rate of strain tensor  $\nabla[\vec{u}]$  dotted with the vorticity vector  $\vec{\omega}$ , where the rate of strain tensor is approximated as

$$\frac{\partial \vec{u}_l}{\partial x_k} = \frac{1}{4\pi} \sum_i \frac{3(\vec{x}(t) - \vec{x}_i(t))_k}{\|\vec{x}(t) - \vec{x}_i(t)\|^5} [(\vec{x}(t) - \vec{x}_i(t)) \times \vec{\alpha}_i(t)]_l - \frac{1}{\|\vec{x}(t) - \vec{x}_i(t)\|^3} [\vec{e}_k \times \vec{\alpha}_i(t)]_l \quad (9)$$

For a regularized representation of the vorticity field,

$$\vec{\omega}_\sigma(\vec{x}, t) = \zeta_\sigma(\vec{x}) * \vec{\omega}(\vec{x}, t) = \sum_i \vec{\alpha}_i(t) \zeta_\sigma(\vec{x} - \vec{x}_i), \quad (10)$$

the potential and velocity vector fields are approximated as

$$\vec{\psi}_\sigma(\vec{x}, t) = \sum_i G_\sigma(\vec{x} - \vec{x}_i) \vec{\alpha}_i(t), \text{ and} \quad (11)$$

$$\vec{u}_\sigma(\vec{x}, t) = \nabla \times \vec{\psi}_\sigma = -\sum_i q_\sigma \|\vec{x}(t) - \vec{x}_i(t)\| \frac{(\vec{x}(t) - \vec{x}_i(t))}{\|\vec{x}(t) - \vec{x}_i(t)\|^3} \times \vec{\alpha}_i(t) \quad (12)$$

where  $\zeta_\sigma(r) = \frac{1}{\sigma^3} \zeta\left(\frac{r}{\sigma}\right)$  is a radially symmetric regularization function and  $\sigma$  is its associated smoothing radius.  $G$  solves the Poisson equation  $\nabla^2 G = -\zeta(r)$ ,  $G_\sigma(r) = \frac{1}{\sigma} G\left(\frac{r}{\sigma}\right)$ ,  $q(r) = \int_0^r \zeta(\rho) \rho^2 d\rho$ , and  $q_\sigma(r) = q\left(\frac{r}{\sigma}\right)$  [9].

The corresponding rate of strain tensor is approximated as

$$\begin{aligned} \frac{\partial \vec{u}_l}{\partial x_k} = & \sum_i \left( 3 \frac{q_\sigma(\|\vec{x}(t) - \vec{x}_i(t)\|)}{\|\vec{x}(t) - \vec{x}_i(t)\|^3} - \zeta_\sigma(\|\vec{x}(t) - \vec{x}_i(t)\|) \right) \frac{(\vec{x}(t) - \vec{x}_i(t))_k}{\|\vec{x}(t) - \vec{x}_i(t)\|^2} [\vec{x} - \vec{x}_i]_l \\ & - \frac{q_\sigma(\|\vec{x}(t) - \vec{x}_i(t)\|)}{\|\vec{x}(t) - \vec{x}_i(t)\|^3} [\vec{e}_k \times \vec{\alpha}_i]_l \end{aligned} \quad (13)$$

To accurately simulate three dimensional flows, a very large number of elements is required to resolve the various length scales. Direct evaluation of the velocity and vortex stretching vectors requires  $O(N^2)$  computational operations of all the pairwise interaction in a system of  $N$  elements. The prohibitive cost of the direct evaluation renders the fast multipole method (FMM) an essential tool to execute particle-based simulations in a reasonable amount of time.

### 3. The fast multipole in three dimensions - adaptive scheme

The computational cost of the  $N$  body problem is tremendously reduced by recursively arranging nearby source elements into smaller boxes and using multipole expansion coefficients  $\vec{M}_n^m$  to approximate the far field components of the velocity and vortex stretching vectors [12]. Further reduction is achieved by clustering target elements and using local expansion coefficients  $\vec{L}_n^m$  to approximate the contributions of all distant sources in the computational domain. The expansion is based on spherical harmonics  $Y_n^m$ .

We consider  $s$  vortices with strengths  $\vec{\alpha}_j$  and position vectors  $\vec{Q}_{j-} = (\rho_j, \theta_j, \varphi_j)$ ,  $j = 1 \dots s$  inside the sphere  $D_{Q_-}$  of center  $Q_-$  with  $|\rho_j| < a$  (see Fig. 1, left). Then at any target location  $\vec{P} = (r, \theta, \varphi)$  with  $r > a$ , the vector potential,  $\vec{\psi}_\delta$ , is approximated by the following multipole expansion [6]

$$\vec{\psi}_\delta(\vec{P}, t) \simeq \frac{1}{4\pi} \sum_{n=0}^p \sum_{m=-n}^n \frac{\vec{M}_n^m}{r^{n+1}} Y_n^m(\theta, \varphi) \quad (14)$$

with

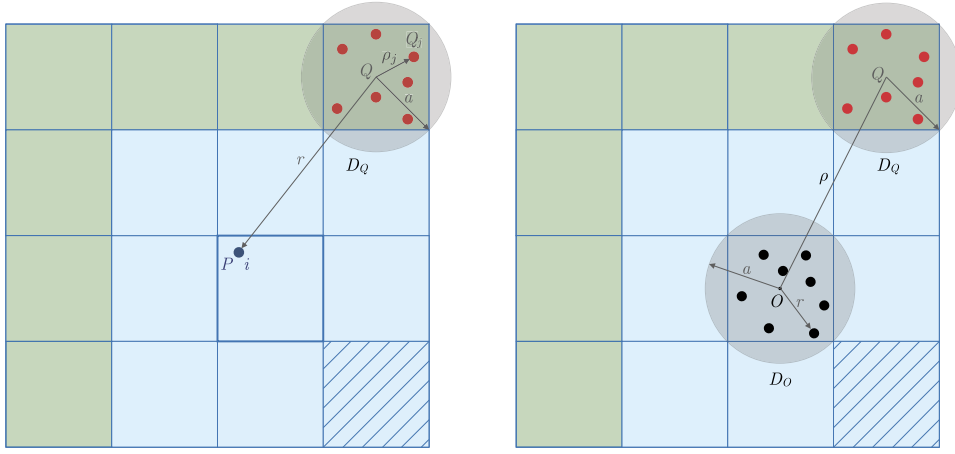


Fig. 1. Schematic for the multipole and local expansions.

$$\vec{M}_n^m = \sum_{j=1}^s \tilde{\alpha}_j \rho_j^n Y_n^{-m}(\theta_j, \varphi_j) \quad (15)$$

where the coordinates of the target position  $\vec{P} = (r, \theta, \varphi)$  are taken with respect to an orthonormal system centered at  $Q$ .

Furthermore, we consider  $s$  vortices having strengths  $(\tilde{\alpha}_j, j = 1 \dots s)$  located inside the sphere  $D_Q$  centered at  $Q = (\rho, \alpha, \beta)$  and of radius  $a$  (Fig. 1, right) such that  $\rho = (c + 1)a$  with  $c > 1$ . Then for any point  $P(r, \theta, \varphi)$  inside sphere  $D_O$  of radius  $a$  centered at the origin, the vector potential,  $\vec{\psi}_\delta$ , due to vortices  $j = 1 \dots s$  is approximated by the following local expansion [6]

$$\vec{\psi}_\delta(P, t) \simeq \frac{1}{4\pi} \sum_{n=0}^p \sum_{m=-n}^n \tilde{L}_n^m Y_n^m(\theta, \varphi) r^n, \quad (16)$$

where

$$\tilde{L}_n^m = \sum_{j=0}^{\infty} \sum_{k=-j}^j \frac{\tilde{M}_j^k \cdot i^{|m-k|-|m|+|k|} \cdot A_n^k \cdot \rho^j \cdot Y_{n+j}^{k-m}(\alpha, \beta)}{(-1)^j A_{n+j}^{k-m} \cdot \rho^{n+j+1}}, \quad (17)$$

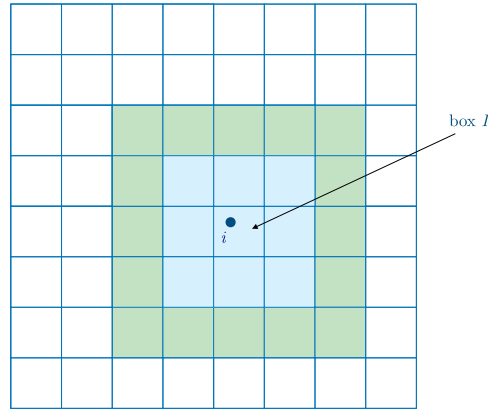
and

$$A_n^m = \frac{(-1)^n}{\sqrt{(n-m)!(n+m)!}} \quad (18)$$

Simple expressions for the velocity and vortex stretching vectors in term of multipole and local expansion coefficients are derived in [7]. The hybrid adaptive error-controlled fast multipole solver introduced in this paper uses these expressions to approximate the far-field vectors.

To facilitate the exposition of the proposed error-controlled adaptive solver, we introduce the following notation:

- $\mathcal{N}_{n_D}(I_l)$ : For a given target box  $I_l$  at level  $l$ , a source box  $J_l$  at the same level belong to the neighborhood of  $I_l$ ,  $\mathcal{N}_{n_D}(I_l)$ , if and only if the distance along the  $x$ ,  $y$ , and  $z$  coordinates between the centers of the two boxes is less than or equal to  $n_D W_{I_l}$  where  $n_D$  is a positive integer and  $W_{I_l}$  is the width of box  $I_l$ . For  $n_D = 1$ ,  $\mathcal{N}_{n_D}(I_l)$  consists of the blue boxes in Fig. 2. For  $n_D = 2$ ,  $\mathcal{N}_{n_D}(I_l)$  consists of the blue and green boxes in Fig. 2.
- $\mathcal{T}_{n_D}(I_l)$ : For a given target box  $I_l$  at level  $l$ , the Interaction List  $\mathcal{T}_{n_D}(I_l)$ , is the set of all boxes which are children of neighbors of  $I_l$ 's parent and are not neighbors of  $I_l$  (green boxes in Fig. 3). For  $n_D = 1$ , the interaction list is similar to that in Cheng et al. (1999) and in the hybrid scheme of Yokota for an optimum MAC value of  $\frac{1}{2}$ .
- $\mathcal{H}_{n_D}(I_l)$ : For a given target box  $I_l$  at level  $l$ , the Inherited List  $\mathcal{H}_{n_D}(I_l)$ , is the set of all boxes that belong to the interaction list of  $I_l$  ancestor, and for which the adaptive solver chooses not to perform cell-cell interaction. As the name suggests, this list is transmitted from a parent box to its children.
- $\mathcal{P}(I_l)$ : Parent of box  $I_l$ .
- $N(I_l)$ : Number of elements in box  $I_l$ . When referring to targets, we use  $N_T(I_l)$ . When referring to sources, we use  $N_S(J_l)$ .
- $isLeaf(I_l)$ : True if  $I_l$  is a leaf box, false otherwise.
- $M(J_l)$ : For a given source box  $J_l$ ,  $M(J_l)$  is the set of multipole expansion coefficients.



**Fig. 2.** Neighborhood  $\mathcal{N}(I_l)$  of box  $I_l$  containing element  $i$ , for  $n_D = 1$  (blue) and  $n_D = 2$  (green). (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

- $L(I_l)$ : For a given target box  $I_l$ ,  $L(I_l)$  is the set of local expansion coefficients.

We note that in our case the target elements are the same as the source elements. The subscripts  $T$  and  $S$  are used to make the discussion more tractable.

The proposed adaptive FMM scheme consists of the following steps:

1. *Construct the root box*

Construct the smallest cubic box that contains all the vortex elements in the computational domain. This box is called the root box.

2. *Build the tree*

Construct a hierarchy of boxes by recursively dividing the root box uniformly into smaller boxes. Refinement level 0 is equivalent to the root box. Refinement level  $l + 1$  is obtained by subdivision of each box at level  $l$  into eight equally sized children. The division process stops according to criterion 1 described in Table 1, where  $n_T^*$  and  $n_F^*$  are defined in section 4 and  $d_\sigma^*$  is defined in section 5.

3. *Initial expansion*

At the finest level, all sources are expanded at their box centers to compute the far-field multipole expansion coefficients  $\bar{M}_n^m$  using equation (15).

4. *Upward pass*

Starting from the leaf boxes (boxes with no children), the multipole expansion coefficients for each source box are recursively translated from the center of the box to the center of the parent box.

5. *Downward pass*

At the center of each box at tree levels  $l \geq 2$ , the local expansion coefficients are recursively calculated. Local expansion coefficients for all boxes at level 1 are first set to zero. For any target box  $I_l$  at level  $l \geq 2$ , initial local expansion coefficients are obtained by translating local coefficients from the center of  $\mathcal{P}(I_l)$  to the center of box  $I_l$ . For any source box  $J_l$  that belongs to the interaction list  $\mathcal{T}_{n_D}(I_l)$ , (green boxes in Fig. 3), We use criterion 2 to decide whether to include  $J_l$  is in the Inherited list of  $I_l$  or convert the multipole expansion coefficients at the center of  $J_l$  into local expansion coefficients at the center of box  $I_l$  using equation (17) and add them to the initial local coefficients. The downward pass stops according to criterion 3.

6. *Compute the velocity and vortex stretching vectors*

For every target element  $i$  in the computational domain, identify leaf box  $I_L$  that contains this element.

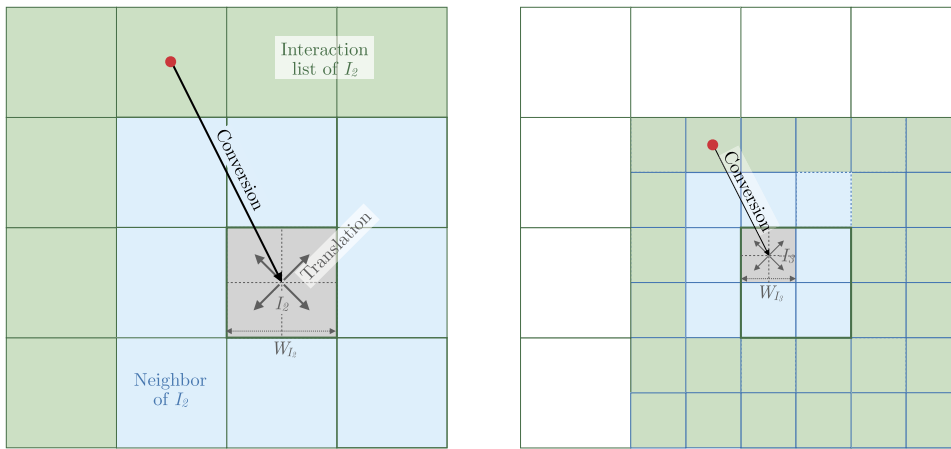
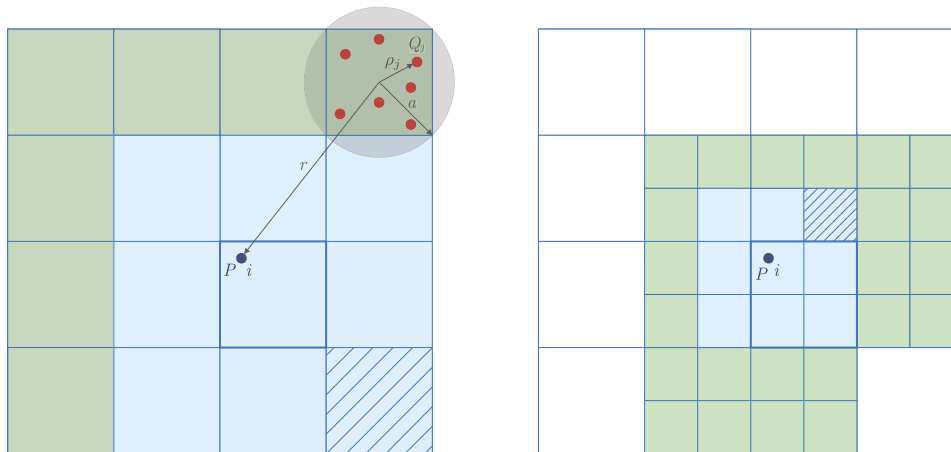
- Calculate the far-field velocity and vortex stretching vectors at the target position using  $I_L$  local expansion coefficients.
- Calculate the velocity and vortex stretching vectors at the target position due to all sources contained within  $I_L$  neighbors and Inherited lists using the recursive method of the  $\mathcal{O}(N \log N)$  scheme (Fig. 4):
  - For each box  $J \in \mathcal{N}_{n_D}(I_L)$  or  $\mathcal{H}_{n_D}(I_L)$ , the velocity and vortex stretching vectors induced at element  $i$  by all source elements contained in  $J$  are calculated by either direct summation or approximated using multipole expansion coefficients according to criterion 4.

The criteria listed in Table 1 are designed to ensure optimal load balancing, while meeting the error threshold requirement. Proper choices of the scheme parameters,  $n_T^*$ ,  $n_F^*$ ,  $d_\sigma^*$ , in addition to the order of expansion  $p$ , are necessary for the proposed scheme to perform as intended. This is carried out through the cost and error analyses discussed next, where we account for the evolving elements distribution, in addition to other factors such as choice of the regularization function and computer architecture.

**Table 1**

Criteria.

| Criterion               | Conditions                                                                                                                                   | Action                                                                   |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Tree building           |                                                                                                                                              |                                                                          |
| 1                       | if $d_{I_l} = d_o^*$ or $N(I_l) < n^*$ ; $n^* = \min\{n_T^*, n_F^*\}$                                                                        | stop division of box $I_l$                                               |
| Downward pass           |                                                                                                                                              |                                                                          |
| 2                       | if $J_l \in \mathcal{T}_{n_D}(I_l)$ and $N_T(I_l) > \frac{n_F^{*2}}{8n_T^*}$<br>and $N_T(I_l) \times N_S(J_l) > \frac{n_F^{*2}}{64}$<br>else | $M(J_l) \rightarrow L(I_l)$<br><br>add $J_l$ to $\mathcal{H}_{n_D}(I_l)$ |
| Computing vector fields |                                                                                                                                              |                                                                          |
| 3                       | if $\text{isLeaf}(I_l)$ or $N_T(I_l) < n_F^*$                                                                                                | stop downward pass                                                       |
| 4                       | if $N_S(J) < \frac{n_T^*}{8}$ or $\text{isLeaf}(J)$<br>else if $i \notin \mathcal{N}_{n_D}(J)$<br>else                                       | direct summation<br>fast summation<br>proceed to next level              |

**Fig. 3.** Schematic for the downward pass in step 5.**Fig. 4.** Schematic for step 6 ( $\mathcal{O}(N \log N)$  FM scheme).

#### 4. Introducing $n_T^*$ and $n_F^*$ through cost analysis

The speed of the adaptive scheme is function of  $p$ ,  $n_D$ , and  $d_m$ , where  $d_m$  is the tree depth. Commonly, the accuracy of the approximation can be improved by increasing  $p$  and/or  $n_D$ , at the expense of a reduction in speed due to the associated additional computational cost. For a given  $p$  and  $n_D$ , the speed of the adaptive scheme increases as  $d_m$  increases up to an

optimal depth,  $d_m^*$ , above which it would be faster and more accurate to perform direct summation than to further divide the tree.

For a given  $p$  and  $n_D$ , let us consider a target box  $I_l$  at level  $l$ . Next we present a criterion, based on the number of elements contained in  $I_l$ , that allows us to decide whether or not it is faster and more accurate to further divide the box.

We denote  $n_T$  as the critical number of elements in box  $I_l$  such that for  $N(I_l) \leq n_T$ , the interaction between  $I_l$  and all sources in neighboring boxes, i.e.  $J_l \in \mathcal{N}_{n_D}(I_l)$ , is evaluated at level  $l$  using direct summation, and for  $N(I_l) > n_T$ , box  $I_l$  is divided into eight children,  $I_{l+1}$ , and the interactions, due to the same sources, are evaluated at level  $l+1$  using both direct summation (for all sources in  $\mathcal{N}_{n_D}(I_{l+1})$ ) and multipole expansion (otherwise). We denote  $n_F$  as the critical number of elements in box  $I_l$  such that for  $N(I_l) \leq n_F$ , the interaction between  $I_l$  and all sources in neighboring boxes, i.e.  $J_l \in \mathcal{N}_{n_D}(I_l)$ , is evaluated at level  $l$  using direct summation, and for  $N(I_l) > n_F$ , box  $I_l$  is divided into eight children,  $I_{l+1}$ , and the interactions, due to the same sources, are evaluated at level  $l+1$  using both direct summation (for all sources within  $\mathcal{N}_{n_D}(I_{l+1})$ ) and local expansion (otherwise). We denote  $n_T^*$  and  $n_F^*$  as the optimal values of  $n_T$  and  $n_F$  respectively. That is, for  $N(I_l) \leq n_T^*$ , dividing this box into eight children and evaluating their influence using multipole expansion coefficients rather than direct summation will lower the speed of the adaptive fast solver. Similarly, for  $N(I_l) \leq n_F^*$ , dividing this box into eight children and evaluating their influence using local expansion coefficients rather than direct summation will lower the speed of the adaptive fast solver. As described in section 3,  $n_T^*$  and  $n_F^*$  are used by the adaptive scheme to optimize the speed of the fast solver.

To facilitate the discussions that follow, we introduce the cost parameters  $a, b, c$  and  $d$ . Let  $a$  be the number of operations needed to calculate the direct interaction at a target  $i$  due to point source  $j$ ,  $b$  the number of operations needed to calculate at a target  $i$  the interaction due to the sources located inside box  $J$  using multipole expansion coefficients,  $c$  the number of operations needed to convert multipole expansion coefficients, calculated at the center of a source box  $J$ , into local expansion coefficients at the center of a distant target box  $I$ , and  $d$  the number of operations needed to translate local expansion coefficients of a target box to one of its children.

#### 4.1. Evaluating $n_T^*$ through cost analysis

The number of operations needed to calculate the direct interaction at target  $i \in I_l$  due to all sources in neighboring boxes, i.e.  $j \in J_l \in \mathcal{N}_{n_D}(I_l)$ , is equal to  $8(n_D + \frac{1}{2})^3 na$ , since there are  $8(n_D + \frac{1}{2})^3$  neighboring boxes containing  $n = \overline{N}_S(J_l)$  sources each.

Now, we estimate, upon division of box  $I_l$  into 8 children, the number of operations required to compute the solution at target  $i$  due to the same sources, i.e.  $j \in J_l \in \mathcal{N}_{n_D}(I_l)$ . Noting that  $i$  belongs to one of the children,  $I_{l+1}$  at level  $l+1$ , the number of operations required to compute the solution at target  $i$  is the sum of direct interactions with sources in  $\mathcal{N}_{n_D}(I_{l+1})$  and fast multipole interaction with sources in  $\mathcal{T}_{n_D}(I_{l+1})$ . The number of operations needed to calculate these interactions are respectively  $(n_D + \frac{1}{2})^3 na$  and  $7 \times 8(n_D + \frac{1}{2})^3 b$ , owing to the fact that there are  $8(n_D + \frac{1}{2})^3$  neighboring boxes, each containing  $\frac{n}{8}$  sources, and  $7 \times 8(n_D + \frac{1}{2})^3$  boxes which are children of box  $I_l$  but are not neighbors of box  $I_{l+1}$ .

The critical number  $n_T^*$  is then equal to the number of elements in box  $I_l$  that yields the same number of operations in the two scenarios considered,

$$8\left(n_D + \frac{1}{2}\right)^3 n_T^* a = \left(n_D + \frac{1}{2}\right)^3 n_T^* a + 7\left(n_D + \frac{1}{2}\right)^3 \times 8 \times b$$

Leading to a value of  $n_T^*$  that is independent of  $n_D$

$$n_T^* = \frac{8b}{a} \quad (19)$$

We conclude that if  $N_T(I_l) > n_T^*$ , it is more cost effective to further divide box  $I_l$  to level  $l+1$  and calculate the velocity and vortex stretching vectors induced by all sources  $j \in \mathcal{N}_{n_D}(I_l)$  by direct interactions with sources in the children's neighborhoods and using multipole expansions with sources in the children's interaction lists. Noting that  $b$  is of the order of  $p^2$ , we may express

$$n_T^* = c_{T2} p^2 + c_{T1} p + c_{T0} \quad (20)$$

where  $c_{T0}$ ,  $c_{T1}$ , and  $c_{T2}$  are functions of the computer system, language, implementation, core function, etc. They also depend on what vector fields are evaluated.

The critical number  $n_T^*$  is used to minimize the number of operations needed to calculate the interactions between a given target  $i \in I_l$  and  $\mathcal{N}_{n_D}(I_l)$ . However, for non-neighboring cells, a different criterion is used by the adaptive solver to optimize the speed of the fast summation. This criterion, which is based on the number of elements in the source box, is used by the adaptive solver to choose between cell-element and element-element interaction. If we consider a box  $J_l$  that contains  $N_S(J_l)$  sources, the number of operations needed to calculate the direct interaction between  $J$  and any target  $i \notin \mathcal{N}_{n_D}(J_l)$  is equal to  $N_S(J_l)a$ . However, since  $i \notin \mathcal{N}_{n_D}(J_l)$ , the interaction between  $J_l$  and  $i$  can alternatively be performed using one Multipole expansion. In this case, it is more cost effective to calculate the velocity and vortex stretching vectors using multipole expansions rather than direct summation if and only if  $N_S(J_l) > \frac{b}{a} = \frac{n_T^*}{8}$ .

#### 4.2. Evaluating $n_F^*$ through cost analysis

The number of operations needed to calculate the direct interaction at target  $i \in I_l$  due to all sources in neighboring boxes, i.e.  $j \in J_l \in \mathcal{N}_{n_D}(I_l)$ , is equal to  $8(n_D + \frac{1}{2})^3 na$ , since there are  $8(n_D + \frac{1}{2})^3$  neighboring boxes containing  $n = \bar{N}_S(J_l)$  sources each. If the number of targets in  $I_l$  is  $N_T(I_l) = n$ , then the number of operations needed to calculate the direct interaction between all targets  $i \in I_l$  and all sources  $j \in \mathcal{N}_{n_D}(I_l)$  is equal to  $8(n_D + \frac{1}{2})^3 n^2 a$ .

Now, we estimate, upon division of box  $I_l$  into 8 children, the number of operations required to compute the solution at all targets  $i \in I_l$  due to the same sources, i.e.  $j \in J_l \in \mathcal{N}_{n_D}(I_l)$ . The number of operations required to compute the solution at all target  $i \in I_{l+1}$  is the sum of direction interactions with sources in  $\mathcal{N}_{n_D}(I_{l+1})$  and the conversion and translation operations. The number of operations needed to calculate the direct interactions is  $(n_D + \frac{1}{2})^3 n^2 a$ , owing to the fact that there are  $8(n_D + \frac{1}{2})^3$  neighboring boxes, each containing  $\frac{n}{8}$  sources. The number of operations needed to perform the conversion operations is equal to  $8 \times 7 \times 8(n_D + \frac{1}{2})^3 c$  since for each of the 8 children,  $I_{l+1}$ , there are  $7 \times 8(n_D + \frac{1}{2})^3$  boxes in  $\mathcal{T}_{n_D}(I_{l+1})$ . In addition, the number of operations needed to perform all the translation operations is equal to  $8d$ .

The critical number  $n_F^*$  can then be estimated by balancing the number of operations in the above two cases

$$8\left(n_D + \frac{1}{2}\right)^3 (n_F^*)^2 a = \left(n_D + \frac{1}{2}\right)^3 (n_F^*)^2 a + 8 \times 7 \times 8\left(n_D + \frac{1}{2}\right)^3 c + 8d$$

leading to

$$n_F^{*2} = 64 \left( \frac{c}{a} + \frac{1}{n_I} \frac{d}{a} \right) \quad (21)$$

where  $n_I = 56(n_D + \frac{1}{2})^3$  is the maximum number of boxes in the interaction list. We conclude that it is more cost effective to further divide box  $I_l$  to level  $l+1$  and then calculate the velocity and vortex stretching vectors using cell-cell interaction if and only if  $N(I_l) > n_F^*$ .

Noting the dependence of  $c$  and  $d$  on  $p^4$ , we may express

$$n_F^{*2} = c_{F4} p^4 + c_{F3} p^3 + c_{F2} p^2 + c_{F1} p + c_{F0} \quad (22)$$

where  $c_{F0}, c_{F1}, c_{F2}, c_{F3}$  and  $c_{F4}$  are functions of the computer system, language, implementation, etc. They also depend on what vector fields are evaluated.

The critical number  $n_F^*$  is used to minimize the number of operations needed to calculate the interactions between a given cell and all its neighbors. However, for non-neighboring cells, a different criterion is used by the adaptive solver to optimize the speed of the fast summation. This criterion is based on the number of elements contained in both interacting and non-neighboring cells and is used by the adaptive solver to choose between cell-cell, cell-element, and element-element interactions. If we consider a source box  $J_l$  that contains  $N_S(J_l)$  elements, the number of operations needed to calculate the direct interaction between  $J_l$  and any target box  $I_l$  outside  $\mathcal{N}_{n_D}(J_l)$  containing  $N_T(I_l)$  elements is equal to  $N_S(J_l) \times N_T(I_l) \times a$ . However, since  $I_l$  is outside  $\mathcal{N}_{n_D}(J_l)$ , the interaction between  $J_l$  and  $I_l$  can be performed using cell-cell interaction with  $c$  operations, and can also be performed using cell-element interactions with  $N_T(I_l) \times b$  operations. In this case, it is more cost effective to calculate the contributions of all sources  $j \in J_l$  at all targets  $i \in I_l, I_l \notin \mathcal{N}_{n_D}(J_l)$  using cell-cell interaction rather than direct summation or cell-element interaction if and only if  $N_S(J_l) \times N_T(I_l) > \frac{c}{a} \simeq \frac{n_F^{*2}}{64}$  and  $N_T(I_l) > \frac{c}{b} \simeq \frac{n_F^{*2}}{8n_T}$ .

For a given architecture, order of expansion  $p$ , and regularization core function, the execution times  $a, b, c, d$  are measured. The values of  $n_T^*$  and  $n_F^*$  are then computed using equations (19) and (21). This procedure is repeated for different values of  $p$ , and the coefficients  $c_{Ti}, i = 0, 1, 2$  and  $c_{Fj}, j = 0, 1, \dots, 4$  are determined using curve fitting.

#### 5. Error analysis of the vector potential and velocity vector for the adaptive scheme with regularized vortices

In the regularized vortex method, where the vorticity field is approximated using equation (10), approximating the far-field vector potential using equations (14) and (16) results in an error made up of three components:

$$\|\vec{E}_{T,\vec{\psi}}\| = \|\vec{E}_{\sigma,\vec{\psi}} + \vec{E}_{M,\vec{\psi}} + \vec{E}_{L,\vec{\psi}}\| \quad (23)$$

where  $\vec{E}_{M,\vec{\psi}}$ ,  $\vec{E}_{L,\vec{\psi}}$ , and  $\vec{E}_{\sigma,\vec{\psi}}$  are respectively the error components due to (i) truncating the multipole expansions at some order  $p$ , (ii) truncating the local expansions at some order  $p$ , and (iii) approximating the Biot-Savart kernel as a  $\frac{1}{r}$  kernel. The multipole expansions truncation component,  $\vec{E}_{M,\vec{\psi}}$ , arises from representing the kernel  $\frac{1}{r}$  at large distance using a series of multipole expansion up to order  $p$ . The local expansions truncation component,  $\vec{E}_{L,\vec{\psi}}$ , arises from representing the kernel  $\frac{1}{r}$  at large distance using a series of local expansion up to order  $p$  and decreases exponentially as  $p$  increases. Both error components,  $\vec{E}_{M,\vec{\psi}}$  and  $\vec{E}_{L,\vec{\psi}}$ , decrease exponentially as  $p$  increases and will eventually converge to zero as

$p \rightarrow \infty$ . Increasing  $p$ , however, increases the computational cost thus reducing the solver speed. We also note that the two truncation errors depend on the Multipole Acceptance Criteria, referred to as  $n_D$ . Increasing  $n_D$  improves the accuracy of the approximations at the expense of a huge reduction in speed arising from the additional cost of the associated direction summation. The error component,  $\tilde{E}_{\sigma, \tilde{\psi}}$ , arises from the fact that  $G_\sigma$  deviates from  $\frac{1}{4\pi r}$ . This error mainly depends on the size of leaf boxes (at the terminal level of the tree). In order to keep  $\tilde{E}_{\sigma, \tilde{\psi}}$  small, the size of the leaf boxes must be larger than a minimum multiple of the core function smoothing radius. In three dimensional flows, this adds considerably to the computational cost of the FM schemes due to the associated increase in element-element interactions.

To estimate upper bounds of these error components, we consider a target element  $i$  in a leaf box  $I_L$ . The adaptive scheme can perform cell-cell, cell-element, and element-element interactions. We denote  $\mathcal{M}$  as the set of all boxes  $J$  at different levels that interact with element  $i$  via cell-element interaction (Multipole Expansion) where the vector potential induced at  $i$  by all vortices contained in  $J$  is calculated using equation (14). We also denote  $\mathcal{L}$  as the set of all boxes at different levels that interact with cell  $I_L$  via cell-cell interaction (Local Expansion) where the vector potential induced at  $i$  by all vortices contained in  $J$  is calculated using equation (16). The upper bounds of the error components associated with computing the vector potential (derivations are presented in Appendix A) are

$$E_{\sigma, \tilde{\psi}} \leq \Gamma_{\mathcal{ML}} \beta_\sigma(r_m) \quad (24)$$

$$\|\tilde{E}_{M, \tilde{\psi}}\| \leq \frac{1}{4\pi} \frac{\Gamma_{\mathcal{M}}}{W_0} \frac{2^{d_m}}{n_D - \frac{\sqrt{3}-1}{2}} \left( \frac{\sqrt{3}}{2n_D + 1} \right)^{p+1} \quad (25)$$

$$\|\tilde{E}_{L, \tilde{\psi}}\| \leq \frac{1}{4\pi} \frac{\Gamma_{\mathcal{L}}}{W_0} \frac{2^{d_m}}{n_D + 1 - \sqrt{3}} \left( \frac{1}{\frac{2\sqrt{3}}{3}(n_D + 1) - 1} \right)^{p+1} \quad (26)$$

where  $\beta(r) = |G(r) - \frac{1}{4\pi r}|$ ,  $\beta_\sigma(r) = \frac{1}{\sigma} \beta(\frac{r}{\sigma})$ ,  $r_m$  is the distance that maximizes the kernel  $\beta_\sigma(r)$ ,  $W_0$  is the width of the root box, and  $\mathcal{ML} = \mathcal{M} \cup \mathcal{L}$ . Furthermore, for any set of boxes  $\mathcal{S}$ , we have:

$$\Gamma_{\mathcal{S}} = \sum_{j \in \mathcal{S}} \|\tilde{\alpha}_j\| \quad (27)$$

The velocity vector is also expressed as a truncation of spherical harmonics. However it has different expressions than those obtained for vector potential, and, as such, has different error bounds. The error in the velocity vector can also be decomposed into three components, namely, the multipole truncation error  $E_{M, \tilde{u}}$ , the local truncation error  $E_{L, \tilde{u}}$  and the regularizing error  $E_{\sigma, \tilde{u}}$ . The total error can be expressed as:

$$\|\tilde{E}_{T, \tilde{u}}\| = \|\tilde{E}_{\sigma, \tilde{u}} + \tilde{E}_{M, \tilde{u}} + \tilde{E}_{L, \tilde{u}}\| \quad (28)$$

The upper bounds of these error components (derivations are presented in Appendix A) are

$$\|\tilde{E}_{\sigma, \tilde{u}}\| \leq \Gamma_{\mathcal{ML}} \kappa_\sigma(r_m) \quad (29)$$

$$\|\tilde{E}_{M, \tilde{u}}\| \leq \frac{1}{4\pi} \frac{\Gamma_{\mathcal{M}}}{W_0^2} \frac{4^{d_m}}{(n_D - \frac{\sqrt{3}-1}{2})^2} \left( \frac{\sqrt{3}}{2n_D + 1} \right)^{p+1} \left[ p + 2 - \frac{\sqrt{3}}{2n_D + 1} (p + 1) \right] \quad (30)$$

$$\|\tilde{E}_{L, \tilde{u}}\| \leq \frac{1}{4\pi} \frac{\Gamma_{\mathcal{L}}}{W_0^2} \frac{4^{d_m}}{(n_D + 1 - \sqrt{3})^2} \left( \frac{1}{\frac{2\sqrt{3}}{3}(n_D + 1) - 1} \right)^{p+1} \left[ p + 2 - \left( \frac{1}{\frac{2\sqrt{3}}{3}(n_D + 1) - 1} \right) p \right] \quad (31)$$

where  $\kappa(r) = \frac{1}{4\pi r^2} |1 - 4\pi q(r)|$  and  $\kappa_\sigma(r) = \frac{1}{\sigma^2} \kappa(\frac{r}{\sigma})$ .

For a given value of  $n_D$ , the truncation errors,  $E_{M, \tilde{u}}$  and  $E_{L, \tilde{u}}$ , mainly depend on the order of expansion  $p$ . For a given required accuracy, there exists an optimal value of the order of expansion,  $p^*$ , below which the truncation errors would be higher than the maximum allowable error. Once calculated,  $p^*$  is used as the truncation order of both multipole and local expansions and to evaluate the two optimal parameters values  $n_T^*$  and  $n_F^*$  using equations (20) and (22) respectively. These optimal parameters values are used by the adaptive solver to limit the depth of the tree in order to balance near-field and far-field evaluations. For a singular (point vortices) representation of the vorticity field where the regularization error vanishes, the division process stops whenever the number of elements in a given box is less than  $n^* = \min\{n_T^*, n_F^*\}$ . In this case,  $n^*$  is the only parameter that controls the depth of the tree structure, i.e.  $d_m^*(n^*)$ . The use of a regularization core function introduces an additional error  $E_{\sigma, \tilde{u}}$  that mainly depends on the maximum level reached upon tree division. In fact, for a given level of accuracy, there exists a critical tree depth  $d_\sigma^*$  above which the regularization error exceeds the maximum allowable error. Thus, in order to maintain the desired accuracy of regularized vortex methods, the tree division must stop whenever the tree level reaches  $d^* = \min(d_\sigma^*, d_m^*(n^*))$ . This additional constraint would impact the FMM balance.

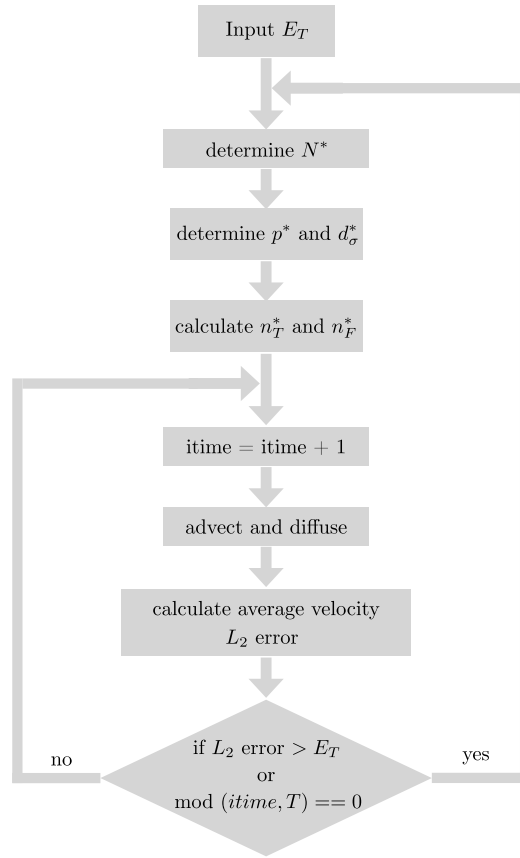


Fig. 5. Flowchart of the error-controlled adaptive scheme.

If  $d_\sigma^* > d_m^*(n^*)$ , the fast solver will manage to balance the loads. On the other hand, if  $d_\sigma^* < d_m^*(n^*)$ , the tree structure would be too shallow and the near field evaluations dominate. In this case, the adaptive solver behaves exactly the same as Greengard FMM [6] since the division process is halted at a given level. However, the adaptive solver will always give the best performance for the desired accuracy and used core function.

Based on the above cost and error discussions, we propose the algorithm presented in Fig. 5 to obtain the best performance for a given accuracy constraint while using a regularized core function. For a given vorticity field representation (elements distribution and core function) and a total error  $E_T$  specified by the user, the adaptive solver first determines the minimum number of samples,  $N^*$ , required to estimate the average  $L_2$  velocity error for a 95% confidence level, and within 5% confidence interval. A sample, of size  $N^*$ , is then used to calculate the value of the optimal order of expansion  $p^*$  below which the Multipole error exceeds the total error, and the critical level  $d_\sigma^*$  above which the regularization error exceeds the total error.  $p^*$  is then used to calculate the values of the optimal parameters  $n_T^*$  and  $n_F^*$  according to equations (20) and (22). Once calculated, these optimal parameters values are used in the adaptive fast solver to balance the far and near-field evaluations, as described in section 3. As the flowfield evolves in times,  $N^*$ ,  $p^*$ ,  $n_T^*$ ,  $n_F^*$ , and  $d_\sigma^*$  are re-evaluated every  $T$  times steps or whenever the calculated error takes a value greater than the pre-defined total error  $E_T$ .

## 6. Results and discussion

### 6.1. Computational cost

The proposed error-controlled algorithm makes use of the two optimal parameters  $n_T^*$  and  $n_F^*$  developed in section 4 to balance the far-field and near-field evaluations for different architectures. For a given architecture, order of expansion  $p$ , and regularizing core function, the execution times  $a, b, c, d$  are measured. The values of  $n_T^*$  and  $n_F^*$  are then computed using equations (19) and (21). This procedure is repeated for different values of  $p$ , and the coefficients  $c_{Ti}$ ,  $i = 0 \dots 2$  and  $c_{Fj}$ ,  $j = 0 \dots 4$  are determined using curve fitting. For example, using an Intel Xeon E5-2620 architecture (2.1 GHz, 8 physical cores) and singular vortices, the dependence of  $n_T^*$  and  $n_F^*$  on  $p$  is shown in Fig. 6 for the case of optimizing the cost of computing the velocity vector, and in Fig. 7 for the case of optimizing the cost of computing both the velocity and vortex stretching vectors. The fitted expressions for  $n_T^*$  and  $n_F^*$  are respectively  $n_{T,\delta}^*(p) = 3.112p^2 + 13.35p + 19.11$ ,  $n_{F,\delta}^*(p) =$

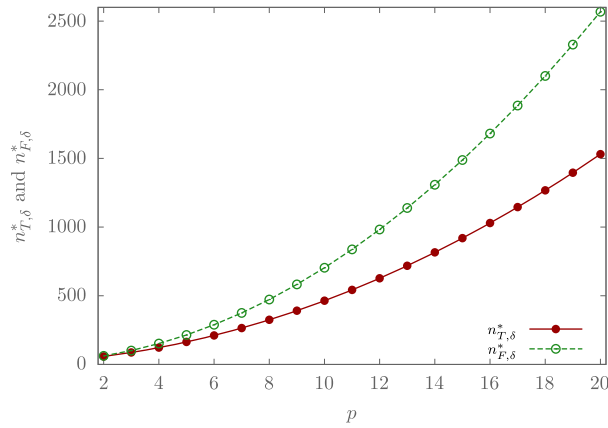


Fig. 6. Optimum  $n_{T,\delta}^*$  and  $n_{F,\delta}^*$  used when calculating only the velocity vector versus  $p$ .

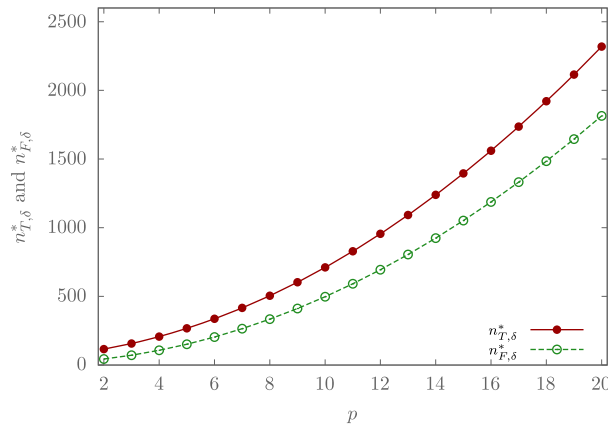


Fig. 7. Optimum  $n_{T,\delta}^*$  and  $n_{F,\delta}^*$  used when calculating both velocity and stretching vectors versus  $p$ .

$5.897p^2 + 9.468p + 19.69$  and  $n_{T,\delta}^*(p) = 4.802p^2 + 16.74p + 63.52$ ,  $n_{F,\delta}^*(p) = 4.167p^2 + 6.689p + 13.91$ , where the subscript  $\delta$  indicates the use of a singular representation of the vorticity field.

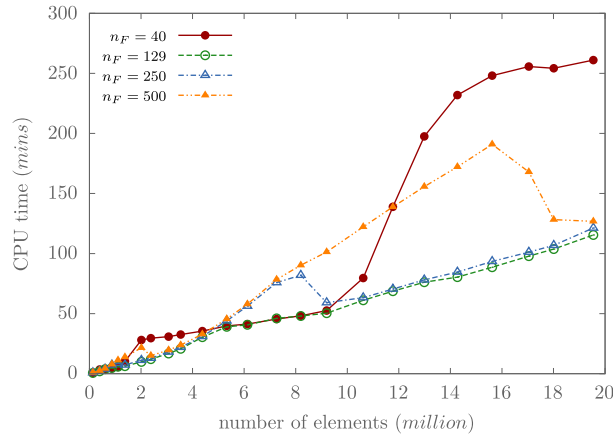
As seen in Figs. 6 and 7,  $n_{T,\delta}^*$  and  $n_{F,\delta}^*$  increase as  $p$  increases. This is due to the fact that it is more expensive to calculate cell-element and cell-cell interactions for larger  $p$ . When only the velocity vector is calculated by the fast scheme,  $n_F^*$  takes larger values than  $n_T^*$ , as shown Fig. 6. Thus further cost reduction can be achieved by reaching higher tree level and using cell-element interaction at these levels to approximate the far-field velocity. At these levels, local expansion coefficients are not calculated since cell-cell interactions would increase the overall computational cost. However when both the velocity and vortex stretching vectors are evaluated by the fast scheme,  $n_F^*$  takes on smaller values than  $n_T^*$  for any order of expansion  $p$ , as shown Fig. 7. Thus local expansion coefficients are calculated for all cells at leaf level and the use of cell-element interaction is the more expensive choice. In this case, the adaptive scheme will basically calibrate the balance between cell-cell and element-element interactions to obtain the optimum speed.

The fact that  $n_{F,\delta}^*$  becomes smaller than  $n_{T,\delta}^*$  when both the velocity and stretching vectors are calculated can be justified as follows. Cell-cell computational cost is mainly due to conversion and translation operations which are used to evaluate local expansions coefficients at the center of each leaf cell. Once obtained, these coefficients are used to approximate the velocity, vortex stretching, and any higher order derivative of the velocity vector field. This is a fixed cost that must be incurred whether we want to calculate only the velocity vector or both the velocity and vortex stretching vectors and it is by far the most expensive component of the cell-cell evaluations. Thus calculating the vortex stretching vector, in addition to velocity vector, would not impact cell-cell computational load since the local expansion coefficients must be calculated either way. Computing the vortex stretching tensor, in addition to the velocity vector, would, however, dramatically increase the element-element and cell-element computational costs. Thus, to achieve the optimum speed,  $n_{F,\delta}^*$  must decrease in order to balance element-element and cell-cell interactions. On the other hand, the corresponding decrease of  $n_{T,\delta}^*$  is rather limited because both element-element and cell-element loads are increasing.

The values of  $n_T^*$  and  $n_F^*$  also depend on the regularization function used to evaluate the near-field velocity and vortex stretching tensor via element-element interaction. For a given regularization core function,  $\zeta(r)$ , the optimal values of  $n_T$  and  $n_F$  are related to  $n_{T,\delta}^*(p)$  and  $n_{F,\delta}^*(p)$  as

**Table 2**  
Value of  $R$  for different core functions.

| Description              | $4\pi\zeta(r)$                                                             | $R$   |
|--------------------------|----------------------------------------------------------------------------|-------|
| Dirac Function           | $\delta(r)$                                                                | 1.000 |
| First order algebraic    | $\frac{3}{(1+r^2)^{\frac{3}{2}}}$                                          | 1.318 |
| Second order algebraic   | $\frac{7.5}{(1+r^2)^{\frac{7}{2}}}$                                        | 1.349 |
| Fourth order algebraic   | $\frac{1}{2} \frac{5-4r^6}{(1+r^6)^{\frac{5}{2}}}$                         | 1.38  |
| Second order Gaussian    | $\sqrt{\frac{2}{\pi}} e^{-\frac{r^2}{2}}$                                  | 1.724 |
| Fourth order Gaussian    | $\sqrt{\frac{2}{\pi}} \left(2.5 - \frac{r^2}{2}\right) e^{-\frac{r^2}{2}}$ | 1.775 |
| Second order Gaussian    | $3e^{-r^3}$                                                                | 1.532 |
| Fourth order exponential | $1.5(5 - 3r^3)e^{-r^3}$                                                    | 1.659 |



**Fig. 8.** Measured simulation time versus the number of elements for different  $n_F$  using an order of expansion  $p = 5$ .

$$n_{T,\sigma}^*(p) = \frac{n_{T,\delta}^*(p)}{R} \quad (32)$$

$$n_{F,\sigma}^*(p) = \frac{n_{F,\delta}^*(p)}{\sqrt{R}} \quad (33)$$

where  $R = \frac{a_\sigma}{a_\delta}$ , and  $a_\delta$  and  $a_\sigma$  are the number of operations needed to calculate the direct interaction between two elements for the case of singular and regularized vortices respectively. Using the same architecture mentioned above, the values of  $R$  for different core functions are listed in Table 2.

To investigate the performance of the adaptive scheme, we carried out several simulations where both velocity and vortex stretching vectors are calculated using a fourth order algebraic core function. A vortex ring with a unit radius and a core radius equal to 0.1 is considered. The initial vorticity within the core of the ring, taken as a second-order Gaussian distribution, is discretized using  $N$  singular vortex elements where  $N$  ranges between 10 thousand and 20 million. Throughout all the simulations,  $n_D$  is set to 1. This is equivalent to a typical fast multipole with a  $3 \times 3 \times 3$  neighbor list. The adaptive scheme employs the criteria listed in Table 1 to perform cell-cell, cell-element, and element-element interactions.

The computational cost, measured in terms of CPU time on a single CPU core, is plotted versus the number of elements in Fig. 8 for  $n_F = 40, 129, 250$ , and  $500$ , using an order of expansions  $p = 5$  and an optimal value of  $n_T^* = 193$ . Inspection of the figure reveals that, for all considered values of the number of elements, the value of  $n_F$  that optimizes the speed of the adaptive scheme by balancing near-field and far-field evaluations is  $n_F = n_{F,\sigma}^*(p) = 129$  where  $p = 5, R = 1.38$ . For other values of  $n_F$ , the adaptive scheme suffers from load imbalance between the near-field and far-field evaluations. Element-element interactions dominate the cost for  $n_F = 250$  and  $500$ , while cell-cell interaction is dominant for the case of  $n_F = 40$ , which is due to the associate cost of computing local expansions for cells in a deep tree structure.

Fig. 8 also shows that as  $n_F$  deviates from its optimum value, the simulation time oscillates above that of  $n_{F,\delta}^* = 129$ . These oscillations assume a higher peak value and a smaller frequency as  $n_F$  further departs from its optimal value. As can be seen in the figure, the load imbalance caused a 250% increase in the CPU time for  $n_F = 50$  and  $N = 20$  million elements.

Fig. 26 in the Appendix shows that similar results are obtained for  $p = 8$ , where the optimal value of  $n_F = n_{F,\sigma}^*(p) = 285$ , chosen by the adaptive solver, gives the best performance.

We conclude that the optimum case has major advantages. By limiting the depth of the tree structure and balancing near-field and far-field evaluation, the adaptive scheme optimizes the speed of the fast multipole method for a given order

of expansion. This scheme is flexible and has auto-tuning capabilities on heterogeneous architectures, and can run on any machine without changing anything. Moreover, the current method will automatically choose the optimal values of the scheme parameters and save the user from this burden. In addition, the adaptive scheme scale as  $\mathcal{O}(N^{1.045})$  and eliminates the number of elements dependence of the method.

## 6.2. Error analysis

For singular representation of the vorticity field, choosing the values of  $n_T^*$  and  $n_F^*$  according to equations (20) and (22) optimizes the speed of the adaptive solver for a given order of expansion  $p$ . In this case, the accuracy of the fast summation is completely determined by the order of expansion. However, the use of a regularization core function induces an additional error, which depends on the used core function,  $\zeta(r)$ , and the neighborhood dimension,  $n_D$ , as discussed in section 5.

To investigate the accuracy of the adaptive scheme, we report the  $L_2$  relative error norms of the velocity and vortex stretching vectors for different values of  $p$  and choices of  $\zeta(r)$ , for the vortex ring considered above, where the initial vorticity distribution is discretized using  $N = 1108880$  elements. For each value of  $p$ , the tree division process stops whenever the number of elements in a box become less than  $n^* = \min\{n_T^*(p), n_F^*(p)\}$  without any constraint on the maximum allowable depth. In this case, the tree depth obtained upon tree division is denoted by  $d_m^*$ , which, for a given elements distribution, is a function of  $n^*$  only. Figs. 9 and 10 show the  $L_2$  relative error norm versus  $p$  for the fourth order algebraic core function. The  $L_2$  relative error norm, which measures the departure of the solution predicted by the adaptive scheme from that obtained by direct summation, is computed using the following expressions

$$L_2(\vec{v}) = \sqrt{\frac{\sum_i \|\vec{v}_a(\vec{x}_i) - \vec{v}_e(\vec{x}_i)\|^2}{\sum_i \|\vec{v}_e(\vec{x}_i)\|^2}} \quad (34)$$

$$L_2(\vec{S}) = \sqrt{\frac{\sum_i \|\vec{S}_a(\vec{x}_i) - \vec{S}_e(\vec{x}_i)\|^2}{\sum_i \|\vec{S}_e(\vec{x}_i)\|^2}} \quad (35)$$

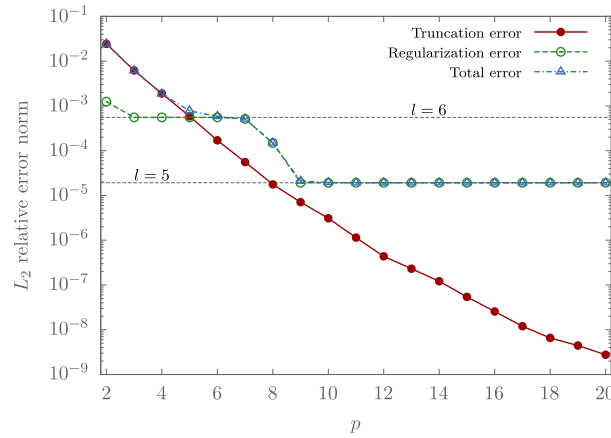
where the subscripts  $e$  and  $a$  refer to the exact direct summation) and approximate solution respectively, and  $\vec{S} = (\vec{\omega} \cdot \nabla) \vec{u}$  is the vortex stretching vector. Figs. 18, 19, 20, and 21 in the Appendix show the  $L_2$  relative error norm of the velocity and vortex stretching vectors plotted against  $p$  for a second order algebraic core function and a second order Gaussian core function respectively. One can observe in Figs. 9, 10, 18, 19, 20 and 21 that there exists a critical value of  $p$  below which the total error is dominated by the truncation error and above which it is dominated by the regularization error. As can be seen in all cases, the truncation error monotonically decreases exponentially, while the regularization error (and thus the total error) decreases in jumps. As  $p$  increases, these jumps occur when the tree depth,  $d_m^*(n^*)$ , decreases (for example from 6 to 5 as shown in the figures) in response to an increase in  $n^*$ . In the range of  $p$  where the regularization error is dominant, increasing  $p$  beyond a certain value  $p^*(d_m^*(n^*))$  (in Figs. 9 and 10,  $p^* = 3$  for  $d_m^*(n^*) = 6$  and 9 for  $d_m^*(n^*) = 5$ ) will only lead to an improvement in accuracy if also accompanied by a drop in the tree depth,  $d_m^*(n^*)$ . This is because the regularization error depends mainly on the core function and on the distance to neighbor cells, which is related to the tree depth  $d_m^*(n^*)$ , which in turn assumes staircase dependence on  $p$ .

Figs. 9, 10, 18, 19, 20 and 21 also show that the  $L_2$  relative error norm of the vortex stretching vector is much larger than that of the velocity vector. In fact, the vortex stretching vector is calculated by taking the derivatives of the velocity vector, expressed as a truncated series of spherical harmonics, during the cell-cell and cell-element operation. As such, the error of the stretching term is expected to be larger than that of the velocity since both vectors are calculated with the same order of expansion  $p$ .

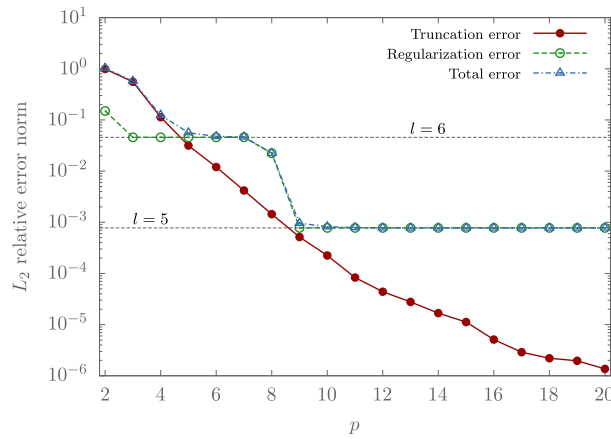
Figs. 11 and 12 show the  $L_2$  relative error norm of the velocity and stretching term vectors versus tree depth,  $d_\sigma$ , for a fixed order of expansion  $p = 5$  using a fourth order algebraic core function. Figs. 22, 23, 24, and 25 in the appendix show the variation of these errors with  $d_\sigma$  for a second order algebraic core function and a second order Gaussian core function, respectively. It is clear that as the tree depth  $d_\sigma$  increases, the regularization error increases while the truncation error remains nearly constant. The increase in the regularization error is because increasing the tree depth  $d_\sigma$  will decrease the distance between neighboring cells and thus increase  $\kappa_\sigma$ .

Figs. 13 and 14 show the relative  $L_2$  regularization error norm versus tree depth,  $d_\sigma$ , for the three core functions considered above and for a fixed order of expansion  $p = 5$ . It can be seen that the regularization error resulting from the use of the fourth order algebraic core function is always smaller than that obtained for a second order algebraic core function. Moreover, the second order Gaussian core function is more accurate than both algebraic functions for small  $d_\sigma$ . However, as  $d_\sigma$  increases, the second order Gaussian core function quickly loses its accuracy to become the least accurate at tree depths larger than 5. Thus, in the low accuracy regime, the second order Gaussian core function is the preferred choice among the core functions considered.

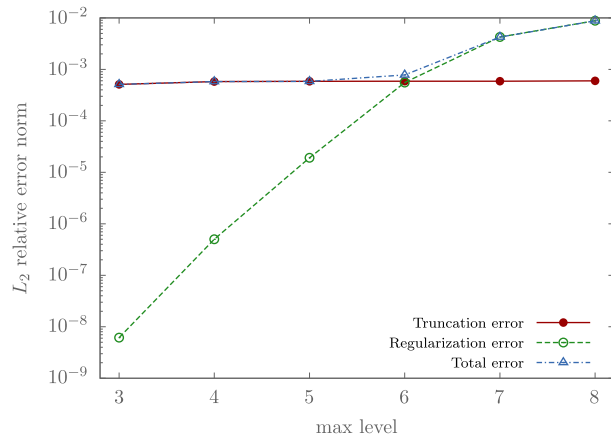
We conclude that, for a regularized representation of the vorticity field, it is inefficient to increase the order of expansion  $p$  unless it is accompanied by a drop in the tree depth since this will only increase the computational cost without improving the accuracy. The proposed hybrid scheme realizes optimal speed for a given accuracy by automatically computing



**Fig. 9.**  $L_2$  relative error norm of the velocity vector versus  $p$  using a fourth order algebraic core function.



**Fig. 10.**  $L_2$  relative error norm of the stretching term vector versus  $p$  using a fourth order algebraic core function.



**Fig. 11.**  $L_2$  relative error norm of the velocity vector versus  $d_\sigma$  for a fixed order of expansion  $p = 5$  using a fourth order algebraic core function.

optimal values of the scheme parameters as follows. The algorithm first determines the optimal tree depth,  $d_\sigma^*$ , beyond which the regularization error exceeds the user defined error threshold. For this optimal level, the algorithm then computes the optimal order of expansion,  $p^*$ , above which increasing  $p$  will only increase the computational cost without improving the total error. As shown in the flowchart of Fig. 5,  $d_\sigma^*$  and  $p^*$  are first computed by evaluating the errors on a representative sample of vortices. Then,  $n_T^*$  and  $n_F^*$  are evaluated as a function of  $p^*$ , using equations (20) and (22).

Next, we present a case study to investigate the efficiency of this method in term of speed and accuracy.

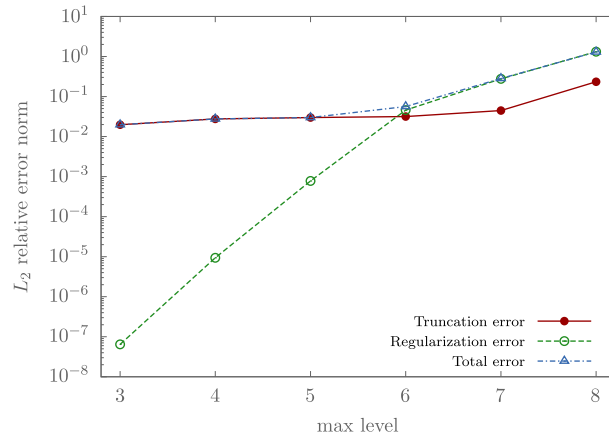


Fig. 12.  $L_2$  relative error norm of the stretching term vector versus  $d_\sigma$  for a fixed order of expansion  $p = 5$  using a fourth order algebraic core function.

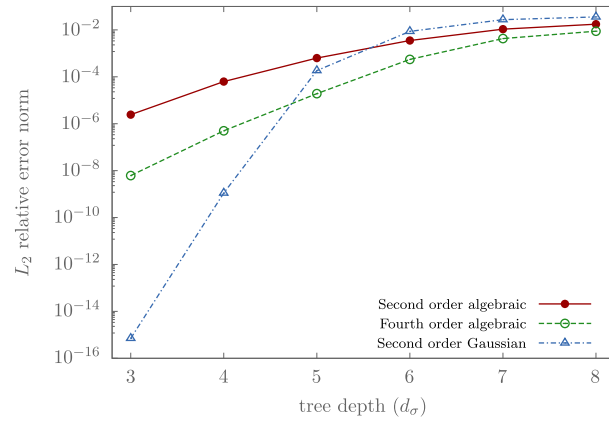


Fig. 13. Relative  $L_2$  regularization error norm of the velocity vector versus  $d_\sigma$  for three different core functions. The order of expansion is  $p = 5$ .

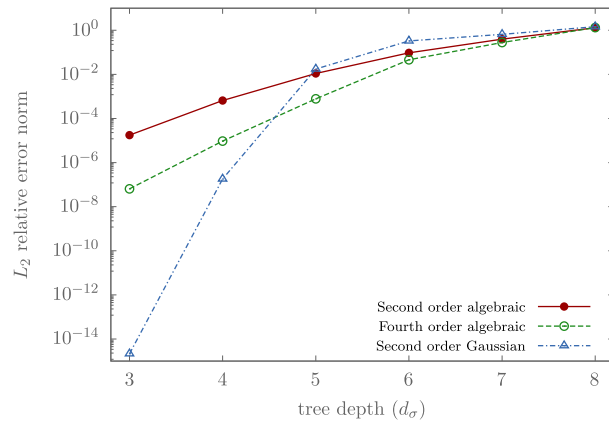


Fig. 14. Relative  $L_2$  regularization error norm of the stretching term vector versus  $d_\sigma$  for three different core functions. The order of expansion is  $p = 5$ .

### 6.3. Case study: collision of two vortex rings

In this section, we explore the performance of the proposed error controlled hybrid adaptive scheme in simulating, using a regularized grid-free vortex method, the collision of two identical axisymmetric vortex rings for Reynolds number  $Re = 1000$ . Initially, each of the two rings is centered on the  $x$  axis and is at an angle of 10 degree from the  $xy$  plane oriented towards the  $z$  axis, as depicted in Fig. 15. The vorticity of the rings, initially along the azimuthal direction, assumes a second-order Gaussian distribution

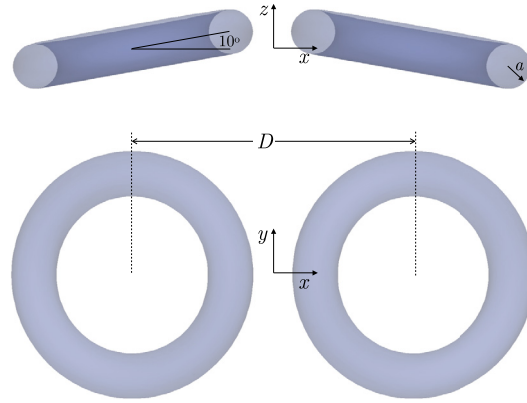


Fig. 15. Setup of the vortex rings at initial time.

$$\vec{\omega}_0(r) = \frac{\Gamma}{2\pi a^2} \exp\left(-\frac{r^2}{2a^2}\right) \vec{e}_\theta \quad (36)$$

where  $r$  is the distance from the center,  $\vec{e}_\theta$  is the unit vector along the azimuthal direction and  $a$  is the dimensionless core radius, taken to be equal to 0.1. The circulation,  $\Gamma$ , around the core is set to unity. The initial vorticity distribution is represented by 159400 equally spaced elements located at the centers of the cells of a uniform radial mesh constructed for each ring. The strength of each particle is set to the product of the value of the vorticity at the particle position and cell volume,  $\vec{\alpha}_i = \vec{\omega}_0(\vec{x}_i) dv$ . The initial vorticity field is approximated by

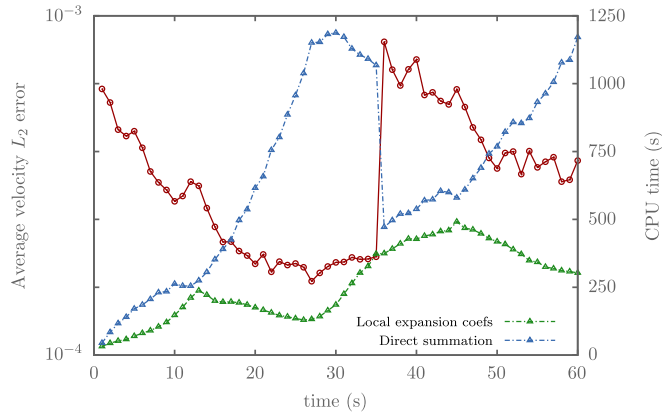
$$\vec{\omega}_\sigma(\vec{x}) = \sum_i \vec{\alpha}_i \zeta_\sigma(\vec{x} - \vec{x}_i), \quad (37)$$

where a fourth order regularizing function with smoothing radius  $\sigma$  is used;  $\zeta_\sigma(\rho) = \frac{1}{\sigma^3} \zeta(\frac{\rho}{\sigma})$ ,  $\zeta(\rho) = \frac{3}{8\pi} \frac{5-4\rho^6}{(1+\rho^6)^{\frac{5}{2}}}$ , and  $\sigma = 2h_0$  where  $h_0 = \sqrt{10\nu\Delta t}$  is the initial elements spacing.

We numerically solve this advection-diffusion problem using a (completely) grid-free vortex method that relies on operator splitting to separately handle advection and diffusion. In the advection step, the elements are convected by the local velocity field using a second order Runge-Kutta time integration scheme with a time step of  $\Delta t = 0.1$  and the strength vector is strained and tilted by the local rate of strain tensor. The velocity and stretching terms are calculated using the adaptive fast multipole algorithm with an order of expansion,  $p^*$ , and tree depth,  $d_\sigma^*$ , calculated by the adaptive solver to meet the accuracy condition specified by the user in terms of a maximum allowable error,  $E_T$ . For the diffusion step, we use the three-dimensional Smooth Redistribution Method (SRM) presented in [7].

As presented in the flowchart of Fig. 5, for a given bounding error  $E_T$ , the adaptive solver first determines the minimum number of samples,  $N^*$ , required to estimate the average velocity error for a 95% confidence level, and within 5% confidence interval. A sample, of size  $N^*$ , is then used to calculate the optimal order of expansion  $p^*$  below which the Multipole error would take higher value than the total error, and the critical tree depth  $d_\sigma^*$  above which the regularization error would exceed the total error. The optimal order of expansion  $p^*$  is used to calculate the two critical numbers  $n_T^*$  and  $n_F^*$  introduced in section 3 and used by the adaptive solver to balance the far and near-field evaluations. This operation is repeated every 50 time steps, or whenever the calculated error takes a value greater than the pre-defined total error  $E_T^* = 10^{-3}$  (see Fig. 5).

In order to assess the efficiency of the fast solver, the CPU time and the average  $L_2$  velocity error are measured every time step, and  $p^*$  and  $d_\sigma^*$  are recorded. Fig. 16 shows, to the left, the average  $L_2$  norm of the velocity error versus time, and to the right, the CPU times for evaluating the local expansion coefficients and direct summation versus time. At  $t = 0$ , the average  $L_2$  norm of the velocity is equal to  $8.84 \times 10^{-4}$  which is smaller than the  $E_T^*$ . The solver automatically satisfies the error bound condition by choosing  $p^* = 4$  and  $d_\sigma^* = 6$ . As time increases, the total error decreases since the average distance between neighboring elements increases which reduces the regularization error. Moreover, the cpu time increases as the number of elements increase. At  $t = 35$ , the optimal order of expansion  $p^*$  dropped from 4 to 3, which immediately resulted in more than 50% savings in the CPU time of the direct evaluation (from 1067 to 475). This is accompanied by a jump in the velocity error from  $3.6 \times 10^{-4}$  to  $8.86 \times 10^{-4}$  which is still smaller than  $E_T^*$ . For  $t > 35$ , the velocity error decreases and the CPU time increases due the increase of the number of elements. The adaptive solver continuously updated, and in an automated fashion, the parameters  $p^*$ ,  $n_F^*$ ,  $n_T^*$  and  $d_\sigma^*$ , to minimize the cost while meeting the accuracy condition, which resulted in considerable savings in the computational cost.



**Fig. 16.** Variation with time of the average velocity  $L_2$  error (left) and the CPU times for computing the local expansion coefficients and the direct summation (right) showing the automated adjustment of  $d_\sigma^*$  and  $p^*$  to reduce the computational cost while meeting the error requirement ( $E_T \leq 10^{-3}$ ). The direct summation cost includes both the velocity and the vortex stretching computations.

## 7. Conclusion

We presented an adaptive error-controlled hybrid fast solver that minimizes, for a given accuracy, the cost of approximating the velocity and vortex stretching vectors using a regularized representation of the vorticity field. The hybrid fast solver uses both multipole and local spherical harmonic based expansions to approximate the vector fields. The proposed solver extends Greengard [6] fast multipole algorithm to continuously optimize the speed of the fast multipole method to adapt to the evolving flow-field, while meeting a given accuracy constraint. The automated scheme employs, in addition to the order of expansion  $p^*$ , the parameters  $n_T^*(p^*)$  and  $n_F^*(p^*)$  to control the depth of the tree structure and to balance the far-field and near-field evaluations by properly choosing, based on well defined criteria, the type of interaction (cell-cell, cell-element, and element-element) to perform in order to compute the contribution of non-neighboring cells. The scheme also uses criteria, based on the critical tree depth  $d_\sigma^*$ , to limit the tree depth to keep the regularization error below the user defined error threshold. Optimal values of the parameters  $p^*$ ,  $n_T^*$ ,  $n_F^*$ , and  $d_\sigma^*$  are periodically evaluated by the automated algorithm to account for the evolving elements distribution. For a given implementation on a given architecture and a given choice of the regularization core function, the dependence of  $n_T^*$  and  $n_F^*$  on  $p$  is established in a one time only preprocessing step.

To assess the performance of the adaptive scheme, we carried out several simulations where the CPU time for evaluating both velocity and vortex stretching vectors is reported for different values of  $n_F$  and different values of the number of elements. These simulations showed that for values of  $n_F$  other than  $n_{F,\sigma}^*(p^*)$ , the adaptive scheme suffers from load imbalance between the near-field and far-field evaluations, resulting in a simulation time that oscillates above that of  $n_{F,\sigma}^*$ , where  $n_{F,\sigma}^* = 129$  for  $p^* = 5$ . Element-element interactions dominate the cost for  $n_F > n_{F,\sigma}^*(p)$ , while cell-cell interaction are dominant for the case of  $n_F < n_{F,\sigma}^*(p)$ , which is due to the associated cost of computing local expansions for cells in a deep tree structure. The simulations also showed that for an optimum value of  $n_F$ , the adaptive scheme scale as  $\mathcal{O}(N^{1.045})$ , hence assuming an almost linear dependence on the number of elements.

To assess the accuracy of the adaptive scheme, we conducted a parametric study to show the impact of the order of expansion,  $p$ , and the maximum tree level,  $d_\sigma$ , on the  $L_2$  relative error norm of the velocity and vortex stretching using different regularization functions. The study revealed that the truncation error exponentially decreases as  $p$  increases, while the regularization error (and thus the total error) decreases in jumps. These jumps occur when the tree depth decreases in response to an increase in  $n^*$ . We concluded that for a regularized representation of the vorticity field, it is inefficient to increase the order of expansion  $p$  unless it is accompanied by a drop in the tree depth since this will only increase the computational cost without improving the accuracy. This implies that the regularization error imposes huge constraints and tremendously decreases the efficiency of the fast multipole methods, which raises a question about the tolerable error for vortex methods. The parametric study also revealed that the  $L_2$  relative error norm of the vortex stretching vector is much higher than that of the velocity vector. Thus, a higher order of expansion is required to calculate the stretching term for a given accuracy, expressed in terms of a bound on the relative  $L_2$  error. This brings forth the important question: is there a way to quantify the relative contributions of the velocity and stretching term errors to the total error induced in solving the Navier-Stokes equation? In fact, the stretching term is used to update the vorticity of each vortex element using the Helmholtz equation, while the velocity is used to advect the vortex elements to their new locations, and there seems to be no “easy” way to quantitatively compare the contributions of these two errors to the total error. This is why, one can control either the velocity error or the stretching term error, but not both. For high Reynolds number simulations, it is crucial that the stretching term is calculated accurately, perhaps even more accurately than the velocity since the stretching term tends to amplify the vorticity divergence. As such, tight control of the error incurred in approximating the stretching term is essential so as to not exacerbate the error in the vorticity divergence.

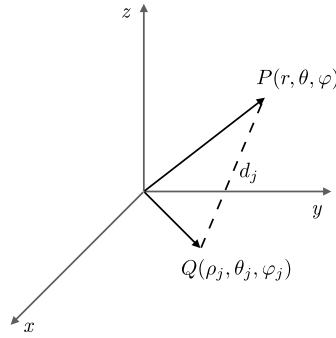


Fig. 17. Schematic showing source and target.

Finally, we showcased the adaptive error-controlled hybrid fast solver by simulating the collision of two vortex rings at Reynolds number  $Re = 1000$  while constraining the maximum error on the average  $L_2$  velocity error to a pre-defined total error of  $E_T = 10^{-3}$ . The results showed that the proposed scheme realizes optimal speed for a given accuracy by automatically computing the optimal values of the scheme parameters to adapt to the evolving flow-field.

### CRediT authorship contribution statement

**Samer Salloum:** Methodology, software, formal analysis, investigation, Writing – Original draft preparation. **Issam Lakkis:** Conceptualization, supervision, software, formal analysis, Writing – Reviewing and Editing.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Data availability

Data will be made available on request.

### Acknowledgements

This work is supported by the University Research Board of the American University of Beirut (project number 25419), URB Award # 103780.

## Appendix A. Error bounds for the potential and velocity vectors

### A.1. Error bounds for the potential vector induced by a singular source element

Consider a singular source element  $j$  of strength  $\vec{\alpha}_j$  located at  $Q = (\rho_j, \theta_j, \varphi_j)$ . The potential vector induced by  $j$  at a target element located at  $P = (r, \theta, \varphi)$  has the following expression:

$$\vec{\psi}_j(P) = \frac{\vec{\alpha}_j}{4\pi d_j}, \quad (38)$$

where  $d_j = \| \vec{QP} \|$  is the distance between source and target elements, as depicted in Fig. 17. Let  $\gamma$  be the angle between the vectors  $P$  and  $Q$ . From the law of cosines, we have:

$$d_j^2 = r^2 + \rho_j^2 - 2r\rho_j \cos\gamma \quad (39)$$

From this relation, we may write

$$\vec{\psi}_j(P) = \frac{\vec{\alpha}_j}{4\pi r} \frac{1}{\sqrt{1 - 2\left(\frac{\rho_j}{r}\right)\cos\gamma + \left(\frac{\rho_j}{r}\right)^2}} \quad (40)$$

Let  $\mu_j = \frac{\rho_j}{r}$ . For  $\rho_j < r$ , that is  $\mu_j < 1$ , if we expand  $\vec{\psi}(P)$  in term of Legendre Polynomial we obtain:

$$\vec{\psi}_j(P) = \frac{\vec{\alpha}_j}{4\pi r} \sum_{n=0}^{\infty} \mu_j^n P_n(\cos\gamma) \quad (41)$$

The error resulting from truncating  $\vec{\psi}_j(P)$  at some order  $p$  is:

$$E_{\vec{\psi}_j} = \left\| \frac{\vec{\alpha}_j}{4\pi d_j} - \frac{\vec{\alpha}_j}{4\pi r} \sum_{n=0}^p \mu_j^n P_n(\cos\gamma) \right\| = \left\| \frac{\vec{\alpha}_j}{4\pi r} \sum_{n=p+1}^{\infty} \mu_j^n P_n(\cos\gamma) \right\| \quad (42)$$

Knowing that  $P_n(\cos\gamma) \leq 1$ , and using the triangle inequality, we obtain an error bound for the multipole expansion

$$E_{\vec{\psi}_j} \leq \left\| \frac{\vec{\alpha}_j}{4\pi r} \sum_{n=p+1}^{\infty} \mu_j^n \right\| \leq \frac{\|\vec{\alpha}_j\|}{4\pi r} \frac{\mu_j^{p+1}}{1 - \mu_j} \quad (43)$$

$$E_{\vec{\psi}_j} \leq \frac{1}{4\pi} \frac{\|\vec{\alpha}_j\|}{(r - \rho_j)} \left( \frac{\rho_j}{r} \right)^{p+1} \quad (44)$$

Similarly, when  $r < \rho_j$ , we define  $\mu_j = \frac{r}{\rho_j}$ . The potential vector and the error bound for the local expansion would have the following expressions:

$$\vec{\psi}_j(P) = \frac{\vec{\alpha}_j}{4\pi d_j} = \frac{\vec{\alpha}_j}{4\pi \rho_j} \sum_{n=0}^{\infty} \mu_j^n P_n(\cos\gamma) \quad (45)$$

$$E_{\vec{\psi}_j} \leq \frac{1}{4\pi} \frac{\|\vec{\alpha}_j\|}{(\rho_j - r)} \left( \frac{r}{\rho_j} \right)^{p+1} \quad (46)$$

#### A.2. Error bounds for the potential vector induced by a cluster of source elements

Suppose that  $s$  vortices with strength  $(\vec{\alpha}_j, j = 1 \dots s)$  are located at the points  $\vec{Q}_j = (\rho_j, \theta_j, \varphi_j)$  inside the sphere  $D_Q$  of radius  $a$  with center at  $Q = (0, 0, 0)$ , then at any  $\vec{P} = (r, \theta, \varphi)$  with  $r > a$ , the potential vector field is given by:

$$\vec{\psi} = \sum_j \vec{\psi}_j = \sum_j \frac{\vec{\alpha}_j}{4\pi d_j} = \sum_j \frac{\vec{\alpha}_j}{4\pi r} \sum_{n=0}^{\infty} \mu_j^n P_n(\cos\gamma) \quad (47)$$

The error resulting from truncating  $\vec{\psi}(P)$  at some order  $p$  is:

$$E_{\vec{\psi}} = \left\| \sum_j \left( \frac{\vec{\alpha}_j}{4\pi d_j} - \frac{\vec{\alpha}_j}{4\pi r} \sum_{n=0}^p \mu_j^n P_n(\cos\gamma) \right) \right\| \quad (48)$$

Using the triangle inequality, and since  $\mu_j \leq \frac{a}{r}$ , the multipole expansion error will be bounded by:

$$E_{\vec{\psi}} \leq \frac{1}{4\pi} \frac{\Gamma_{D_Q}}{(r - a)} \left( \frac{a}{r} \right)^{p+1} \quad (49)$$

where  $\Gamma_{D_Q} = \sum_j \|\vec{\alpha}_j\|$

Now suppose that  $s$  vortices with strength  $(\vec{\alpha}_j, j = 1 \dots s)$  are located inside a cubic box  $J_l$  at level  $l$  with center at  $Q = (0, 0, 0)$ . The radius of the smallest sphere which encloses box  $J_l$  is  $a_{J_l} = \frac{\sqrt{3}}{2} W_{J_l}$  where  $W_{J_l}$  is the box width. Following the adaptive solver Scheme described in section 3, the distance between the center of box  $J_l$  and any point  $P = (r, \theta, \varphi)$  located outside the neighborhood of box  $J_l$  is  $r_{J_l} \geq (n_D + 0.5)W_{J_l}$ . For any  $p \geq 1$ , the multipole expansion error will be bounded by:

$$\bar{E}_{M, \vec{\psi}} \leq \frac{1}{4\pi} \frac{\Gamma_{J_l}}{W_{J_l}} \frac{1}{n_D - \frac{\sqrt{3}-1}{2}} \left( \frac{\sqrt{3}}{2n_D + 1} \right)^{p+1} \quad (50)$$

Noting that  $W_{J_l} = W_0/2^l$ , we get

$$E_{M, \vec{\psi}} \leq \frac{1}{4\pi} \frac{\Gamma_{J_l}}{W_0} \frac{2^l}{n_D - \frac{\sqrt{3}-1}{2}} \left( \frac{\sqrt{3}}{2n_D + 1} \right)^{p+1} \quad (51)$$

where  $W_0$  is the width of the root box,  $\Gamma_{J_l} = \sum_j \|\vec{\alpha}_j\|$  and the summation is done over all the source elements within box  $J_l$ .

Similarly, suppose that  $s$  vortices with strength  $(\vec{\alpha}_i, i = 1 \dots s)$  are located inside the sphere  $D_Q$  of radius  $a$  with center at  $Q = (\rho, \alpha, \beta)$ , and that  $\rho = (c+1)a$  with  $c > 1$ , then for any target  $P = (r, \theta, \phi)$  inside the sphere  $D_0$  of radius  $a$  centred at the Origin, the potential vector is given by

$$\vec{\psi} = \sum_j \vec{\psi}_j = \sum_j \frac{\vec{\alpha}_j}{4\pi d_j} = \sum_j \frac{\vec{\alpha}_j}{4\pi \rho_j} \sum_{n=0}^{\infty} \mu_j^n P_n(\cos\gamma) \quad (52)$$

The error resulting from truncating  $\vec{\psi}(P)$  at some order  $p$  is:

$$E_{\vec{\psi}} = \left\| \sum_j \left( \frac{\vec{\alpha}_j}{4\pi d_j} - \frac{\vec{\alpha}_j}{4\pi \rho_j} \sum_{n=0}^p \mu_j^n P_n(\cos\gamma) \right) \right\| \quad (53)$$

Since  $\mu_j = \frac{r}{\rho_j} \leq \frac{a}{\rho-a} = \frac{1}{c}$ , using the triangle inequality, the local expansion error will be bounded by:

$$E_{\vec{\psi}} \leq \frac{1}{4\pi} \frac{\Gamma_{D_Q}}{(ca-a)} \left( \frac{1}{c} \right)^{p+1} \quad (54)$$

where  $\Gamma_{D_Q} = \sum_j \|\vec{\alpha}_j\|$ .

Now suppose that  $s$  vortices with strength  $(\vec{\alpha}_j, j = 1 \dots s)$  are located inside a cubic box  $J_l$  at level  $l$  with center at  $Q = (\rho, \alpha, \beta)$ . The radius of the smallest sphere which encloses box  $J_l$  is  $a_{J_l} = \frac{\sqrt{3}}{2} W_{J_l}$  where  $W_{J_l}$  is the box width. Following the adaptive solver Scheme described in section 3, an upper bound for  $c$  is given by

$$c \geq \frac{2\sqrt{3}}{3} (n_D + 1) - 1, \quad (55)$$

and for any  $p \geq 1$ , the local expansion error that results upon approximating the potential vector at any point  $P$  within a box  $I_l$  centered at the origin is bounded by:

$$E_{L, \vec{\psi}} \leq \frac{1}{4\pi} \frac{\Gamma_{J_l}}{W_0} \frac{2_l}{n_D + 1 - \frac{\sqrt{3}}{2}} \left( \frac{1}{\frac{2\sqrt{3}}{3} (n_D + 1) - 1} \right)^{p+1} \quad (56)$$

### A.3. Error bounds for the velocity vector induced by a singular source element

Once again, let us consider a singular source element  $j$  of strength  $\vec{\alpha}_j$  located at  $Q = (\rho_j, \theta_j, \varphi_j)$ . The velocity at a target element  $P = (r, \theta, \varphi)$  is calculated as the curl of the potential vector  $\vec{\psi}_j$

$$\vec{u}_j(P) = \nabla \times \vec{\psi}_j(P) = \nabla \times \left( \frac{\vec{\alpha}_j}{4\pi r} \sum_{n=0}^{\infty} \mu_j^n P_n(\cos\gamma) \right) = \sum_{n=0}^{\infty} \frac{\rho_j^n}{4\pi} \nabla \left( \frac{P_n(\cos\gamma)}{r^{n+1}} \right) \times \vec{\alpha}_j \quad (57)$$

The error resulting from truncating  $\vec{u}_j(P)$  at some order  $p$  is:

$$E_{\vec{u}_j} = \left\| \sum_{n=p+1}^{\infty} \frac{\rho_j^n}{4\pi} \nabla \left( \frac{P_n(\cos\gamma)}{r^{n+1}} \right) \times \vec{\alpha}_j \right\| \quad (58)$$

Since  $\|\vec{u} \times \vec{v}\| \leq \|\vec{u}\| \|\vec{v}\|$  for any two vectors  $\vec{u}$  and  $\vec{v}$ , using the triangle inequality, the truncating error will be bounded by:

$$E_{\vec{u}_j} \leq \frac{\|\vec{\alpha}_j\|}{4\pi} \sum_{n=p+1}^{\infty} \rho_j^n \left\| \nabla \left( \frac{P_n(\cos\gamma)}{r^{n+1}} \right) \right\| \quad (59)$$

Since the norm of a vector does not vary when we change the reference frame, we will evaluate the expression of  $\left\| \nabla \left( \frac{P_n(\cos\gamma)}{r^{n+1}} \right) \right\|$  in a new frame obtained by rotating the old frame such that the source element  $Q$  will be located on the new  $zz'$  axis, that is we have  $\cos\gamma = \cos\theta$

$$\nabla \left( \frac{P_n(\cos\theta)}{r^{n+1}} \right) = -\frac{1}{r^{n+2}} ((n+1) P_n(\cos\theta), \sin(\theta) P'_n(\cos\theta), 0)^T \quad (60)$$

$$\left\| \nabla \left( \frac{P_n(\cos\theta)}{r^{n+1}} \right) \right\| = \frac{1}{r^{n+2}} \sqrt{(n+1)^2 P_n^2(\cos\theta) + \sin^2(\theta) P_n'^2(\cos\theta)} \quad (61)$$

Knowing that

$$(n+1)^2 P_n^2(x) + (1-x^2) P_n'^2(x) \leq (n+1)^2 \quad (62)$$

we obtain:

$$\left\| \nabla \left( \frac{P_n(\cos\theta)}{r^{n+1}} \right) \right\| \leq \frac{n+1}{r^{n+2}} \quad (63)$$

Replacing in equation (59), the multipole expansion error of the velocity vector is bounded by:

$$E_{\vec{u}_j} \leq \frac{\|\vec{\alpha}_j\|}{4\pi r^2} \sum_{n=p+1}^{\infty} (n+1) \mu_j^n \quad (64)$$

$$E_{\vec{u}_j} \leq \frac{\|\vec{\alpha}_j\|}{4\pi r^2} \left[ \frac{\mu_j^{p+1}}{1-\mu_j} + \mu_j \frac{d}{d\mu_j} \left( \frac{\mu_j^{p+1}}{1-\mu_j} \right) \right] \quad (65)$$

$$E_{\vec{u}_j} \leq \frac{\|\vec{\alpha}_j\|}{4\pi r^2} \frac{\mu_j^{p+1}}{(1-\mu_j)^2} [p+2 - \mu_j(p+1)] \quad (66)$$

$$E_{\vec{u}_j} \leq \frac{1}{4\pi} \frac{\|\vec{\alpha}_j\|}{(r-\rho_j)^2} \left( \frac{\rho_j}{r} \right)^{p+1} \left[ p+2 - \frac{\rho_j}{r} (p+1) \right] \quad (67)$$

Similarly, when  $r < \rho_j$ ,  $\mu_j = \frac{r}{\rho_j}$ , the velocity vector induced by  $j$  at a target  $P = (r, \theta, \phi)$  is calculated as the curl of the potential vector and has the following expression:

$$\vec{u}_j(P) = \nabla \times \vec{\psi}_j(P) = \nabla \times \left( \frac{\vec{\alpha}_j}{4\pi \rho_j} \sum_{n=0}^{\infty} \mu_j^n P_n(\cos\gamma) \right) = \sum_{n=0}^{\infty} \frac{1}{4\pi \rho_j^n} \nabla (r^n P_n(\cos\gamma)) \times \vec{\alpha}_j \quad (68)$$

The error resulting from truncating  $\vec{u}_j(P)$  at some order  $p$  is:

$$E_{\vec{u}_j} = \left\| \sum_{n=p+1}^{\infty} \frac{1}{4\pi \rho_j^n} \nabla (r^n P_n(\cos\gamma)) \times \vec{\alpha}_j \right\| \quad (69)$$

$$E_{\vec{u}_j} \leq \frac{\|\vec{\alpha}_j\|}{4\pi} \sum_{n=p+1}^{\infty} \frac{1}{\rho_j^n} \|\nabla (r^n P_n(\cos\gamma))\| \quad (70)$$

Once again, the norm of a vector does not vary when we change the reference frame. Calculating the norm in the new reference frame described above, we obtain:

$$\nabla (r^n P_n(\cos\gamma)) = r^{n-1} (n P_n(\cos\theta), -\sin\theta P_n'(\cos\theta), 0)^T \quad (71)$$

$$\|r^n P_n(\cos\gamma)\| = r^{n-1} \sqrt{n^2 P_n^2(\cos\theta) + \sin^2\theta P_n'^2(\cos\theta)} \quad (72)$$

Knowing that

$$n^2 P_n^2(x) + (1-x^2) P_n'^2(x) \leq n^2 \quad (73)$$

we obtain:

$$\|r^n P_n(\cos\gamma)\| = n r^{n-1} \quad (74)$$

Replacing in equation (70), the local expansion error of the velocity vector is bounded by:

$$E_{\vec{u}_j} \leq \frac{\|\vec{\alpha}_j\|}{4\pi \rho_j} \sum_{n=p+1}^{\infty} n \mu_j^n \quad (75)$$

$$E_{\vec{u}_j} \leq \frac{\|\vec{\alpha}_j\|}{4\pi \rho_j} \mu_j \frac{d}{d\mu_j} \left( \frac{\mu_j^{p+1}}{1-\mu_j} \right) \quad (76)$$

$$E_{\tilde{u}_j} \leq \frac{\|\tilde{\alpha}_j\|}{4\pi\rho_j^2} \frac{\mu_j^{p+1}}{(1-\mu_j)^2} [p+2-\mu_j p] \quad (77)$$

$$E_{\tilde{u}_j} \leq \frac{1}{4\pi} \frac{\|\tilde{\alpha}_j\|}{(\rho_j-r)^2} \left(\frac{r}{\rho_j}\right)^{p+1} \left[p+2-\frac{r}{\rho_j}p\right] \quad (78)$$

#### A.4. Error bounds for the velocity vector induced by a cluster of source elements

Suppose that  $s$  vortices with strength  $(\tilde{\alpha}_j, j=1 \dots s)$  are located at the points  $\tilde{Q}_j = (\rho_j, \theta_j, \varphi_j)$  inside the sphere  $D_Q$  of radius  $a$  with center at  $Q = (0, 0, 0)$ , then at any  $\tilde{P} = (r, \theta, \varphi)$  with  $r > a$ , the velocity vector field is given by:

$$\tilde{u} = \sum_j \tilde{u}_j = \sum_j \sum_{n=0}^{\infty} \frac{\rho_j^n}{4\pi} \nabla \left( \frac{P_n(\cos\gamma)}{r^{n+1}} \right) \times \tilde{\alpha}_j \quad (79)$$

Using the triangle inequality, and since  $\mu_j \leq \frac{a}{r}$ , the multipole expansion error will be bounded by:

$$E_{\tilde{u}} \leq \sum_i E_{\tilde{u}_i} \leq \frac{1}{4\pi} \frac{\Gamma_{D_Q}}{(r-a)^2} \left(\frac{a}{r}\right)^{p+1} \left[p+2-\frac{a}{r}(p+1)\right] \quad (80)$$

where  $\Gamma_{D_Q} = \sum_j \|\tilde{\alpha}_j\|$ .

Now suppose that  $s$  vortices with strength  $(\tilde{\alpha}_j, j=1 \dots s)$  are located inside a cubic box  $J_l$  at level  $l$  with center at  $Q = (0, 0, 0)$ . For any  $p \geq 1$ , the multipole expansion error that results upon approximating the velocity vector at any point  $P$  outside the neighbors of  $J_l$  is bounded by:

$$E_{M,\tilde{u}} \leq \frac{1}{4\pi} \frac{\Gamma_{J_l}}{W_0^2} \frac{4^l}{\left(n_D - \frac{\sqrt{3}-1}{2}\right)^2} \left(\frac{\sqrt{3}}{2n_D+1}\right)^{p+1} \left[p+2-\frac{\sqrt{3}}{2n_D+1}(p+1)\right] \quad (81)$$

Similarly, suppose that  $s$  vortices having circulation  $(\tilde{\alpha}_i, i=1 \dots s)$  are located inside the sphere  $D_Q$  of radius  $a$  with center at  $Q = (\rho, \alpha, \beta)$ , and that  $\rho = (c+1)a$  with  $c > 1$ , then for any target  $P = (r, \theta, \phi)$  inside the sphere  $D_0$  of radius  $a$  centred at the Origin, the velocity vector is given by

$$\tilde{u} = \sum_j \tilde{u}_j = \sum_j \sum_{n=0}^{\infty} \frac{1}{4\pi\rho_j^n} \nabla (r^n P_n(\cos\gamma)) \times \tilde{\alpha}_j \quad (82)$$

Since  $\mu_j = \frac{r}{\rho_j} \leq \frac{a}{\rho-a} = \frac{1}{c}$ , using the triangle inequality, the local expansion error will be bounded by:

$$E_{\tilde{u}} \leq \sum E_{\tilde{u}_j} \leq \frac{1}{4\pi} \frac{\Gamma_{D_Q}}{(ca-c)^2} \left(\frac{1}{c}\right)^{p+1} \left[p+2-\frac{1}{c}p\right] \quad (83)$$

where  $\Gamma_{D_Q} = \sum_j \|\tilde{\alpha}_j\|$ .

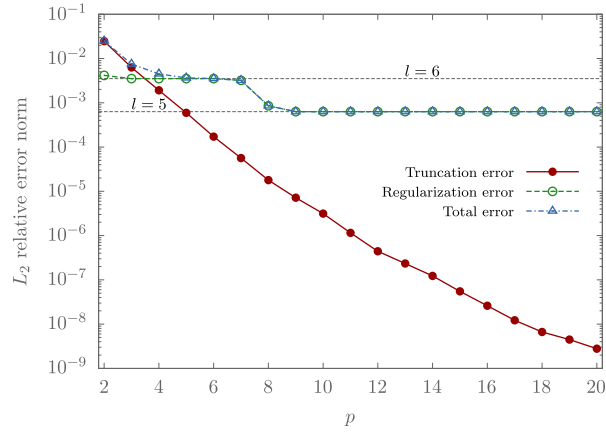
Now suppose that  $s$  vortices with strength  $(\tilde{\alpha}_j, j=1 \dots s)$  are located inside a cubic box  $J_l$  at level  $l$  with center at  $Q = (\rho, \alpha, \beta)$ . For any  $p \geq 1$ , the local expansion error that results upon approximating the velocity vector at any point  $P$  within a box  $I_l$  centered at the origin is bounded by:

$$E_{L,\tilde{u}} \leq \frac{1}{4\pi} \frac{\Gamma_{J_l}}{W_0^2} \frac{4^l}{\left(n_D+1-\sqrt{3}\right)^2} \left(\frac{1}{\frac{2\sqrt{3}}{3}(n_D+1)-1}\right)^{p+1} \left[p+2-\left(\frac{1}{\frac{2\sqrt{3}}{3}(n_D+1)-1}\right)p\right] \quad (84)$$

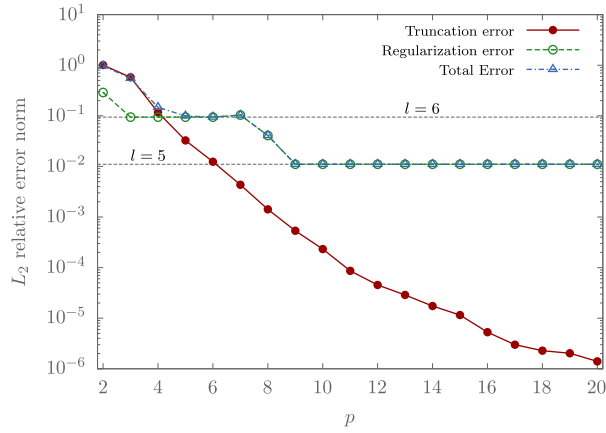
#### A.5. Regularization error bounds for the potential vector

In the case of regularized vortex method, an additional error  $E_{\sigma,\tilde{\psi}}$  arises since  $G_\sigma$  deviates from  $\frac{1}{4\pi r}$ . Let us consider  $s$  vortices with strength  $(\tilde{\alpha}_j, j=1 \dots s)$  located inside a cubic box  $J_l$ . Introducing the kernel  $\beta_\sigma(r) = \frac{1}{\sigma} \beta\left(\frac{r}{\sigma}\right)$ , and  $\beta(r) = |G(r) - \frac{1}{4\pi r}|$  we obtain:

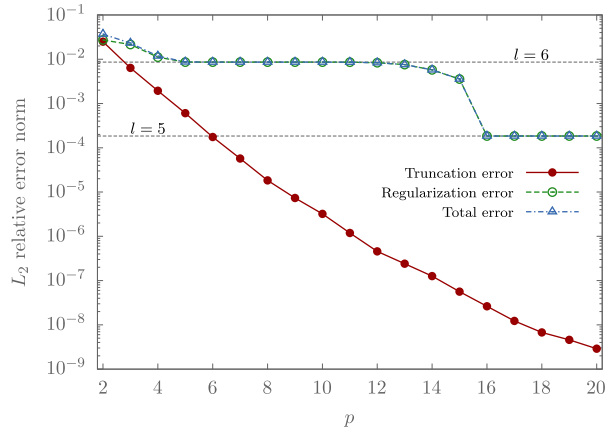
$$\begin{aligned} E_{\sigma,\tilde{\psi}} &= \left\| \sum_{j \in J_l} \tilde{\alpha}_j \left( G_\sigma(r_j) - \frac{1}{4\pi r_j} \right) \right\| \\ &\leq \sum_{j \in J_l} \|\tilde{\alpha}_j\| \left| G_\sigma(r_j) - \frac{1}{4\pi r_j} \right| \end{aligned}$$



**Fig. 18.**  $L_2$  relative error norm of the velocity vector versus  $p$  using a second order algebraic core function.



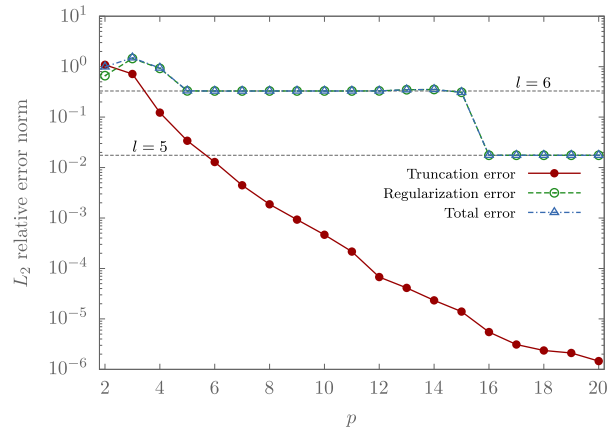
**Fig. 19.**  $L_2$  relative error norm of the stretching term vector versus  $p$  using a second order algebraic core function.



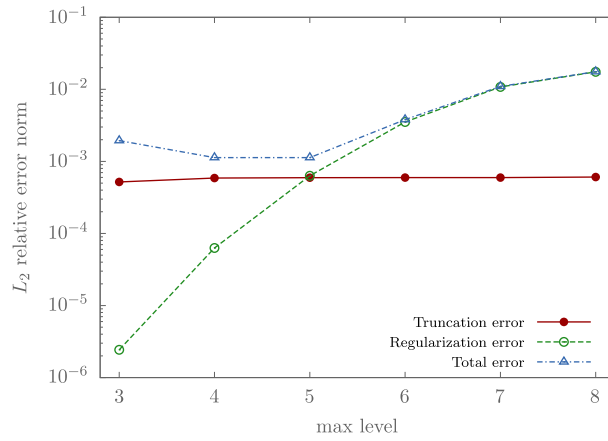
**Fig. 20.**  $L_2$  relative error norm of the velocity vector versus  $p$  using a second order Gaussian core function.

$$\begin{aligned}
 &\leq |G_\sigma(r_m) - \frac{1}{4\pi r_m}| \sum_{j \in J_l} \|\tilde{\alpha}_j\| \\
 &\leq \Gamma_{J_l} \beta_\sigma(r_m)
 \end{aligned} \tag{85}$$

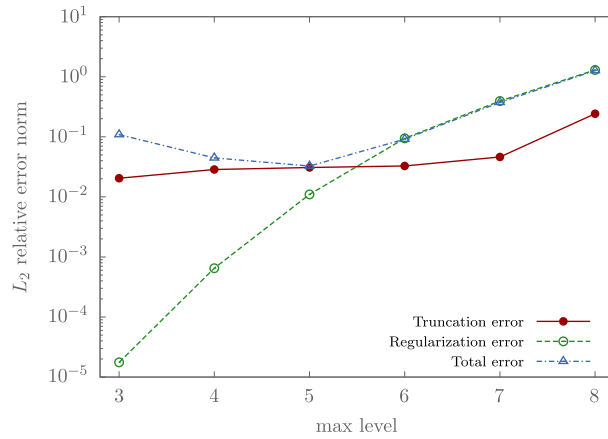
where  $\Gamma_{J_l} = \sum_j \|\tilde{\alpha}_j\|$ ,  $r_j$  is the distance between source element  $j$  and  $P$ ,  $r_m$  is the distance that maximizes the kernel  $\beta_\sigma(r)$ .  $r^*$  depends only on the choice of the core function and for most cases we have  $r_m = (n_D + 0.5)W_{min} = (n_D + 0.5)\frac{W_0}{2}$ .



**Fig. 21.**  $L_2$  relative error norm of the stretching term vector versus  $p$  using a second order Gaussian core function.



**Fig. 22.**  $L_2$  relative error norm of the velocity vector versus  $d_\sigma$  for a fixed order of expansion  $p = 5$  using a second order algebraic core function.



**Fig. 23.**  $L_2$  relative error norm of the stretching term vector versus  $d_\sigma$  for a fixed order of expansion  $p = 5$  using a second order algebraic core function.

#### A.6. Regularization error bounds for the velocity vector

The regularizing error for the velocity arose since  $q_\sigma(r)$  deviates from  $\frac{1}{4\pi}$ . Let us consider  $s$  vortices with strength  $(\vec{\alpha}_j, j = 1 \dots s)$  located inside a cubic box  $J_l$ . For any target element  $i$  we have:

$$E_{\sigma, \vec{u}} = \left\| \sum_{j \in J_l} \frac{1}{4\pi r_j^3} (\vec{x}_i - \vec{x}_j) \wedge \vec{\alpha}_j - \frac{q_\sigma(r_j)}{r_j^3} (\vec{x}_i - \vec{x}_j) \wedge \vec{\alpha}_j \right\|$$

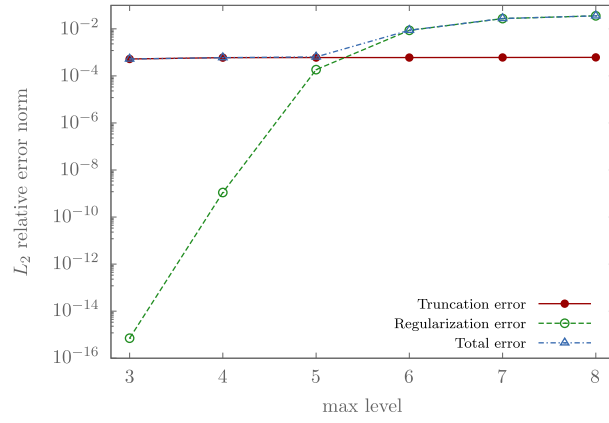


Fig. 24.  $L_2$  relative error norm of the velocity vector versus  $d_\sigma$  for a fixed order of expansion  $p = 5$  using a second order Gaussian core function.

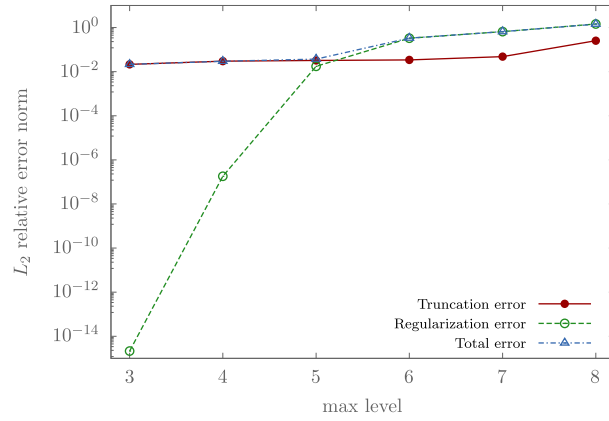


Fig. 25.  $L_2$  relative error norm of the stretching term vector versus  $d_\sigma$  for a fixed order of expansion  $p = 5$  using a second order Gaussian core function.

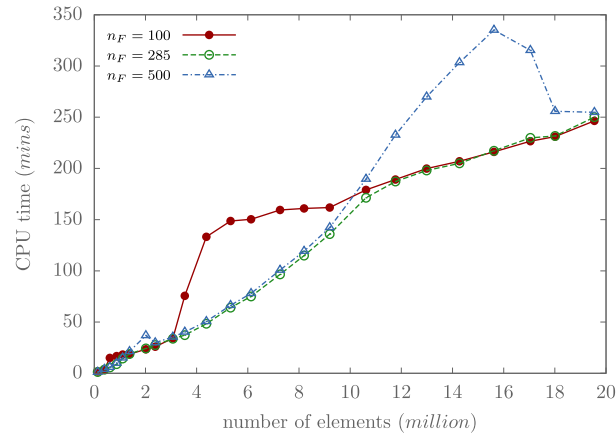


Fig. 26. Measured simulation time versus the number of elements for different  $n_F$  using an order of expansion  $p = 8$ .

$$\begin{aligned}
 &\leq \sum_{j \in J_l} \left\| \frac{1}{4\pi r_j^3} (\vec{x}_i - \vec{x}_j) \wedge \vec{\alpha}_j - \frac{q_\sigma(r_j)}{r_j^3} (\vec{x}_i - \vec{x}_j) \wedge \vec{\alpha}_j \right\| \\
 &\leq \sum_{j \in J_l} \frac{1}{4\pi r_j^3} |1 - 4\pi q_\sigma(r_j)| \left\| (\vec{x}_i - \vec{x}_j) \wedge \vec{\alpha}_j \right\| \\
 &\leq \sum_{j \in J_l} \frac{1}{4\pi r_j^2} |1 - 4\pi q_\sigma(r_j)| \left\| \vec{\alpha}_j \right\|
 \end{aligned}$$

$$\begin{aligned}
&\leq \frac{\Gamma_{J_l}}{4\pi r_m^2} |1 - 4\pi q(\frac{r_m}{\sigma})| \\
&\leq \Gamma_{J_l} \kappa_\sigma(r_m)
\end{aligned} \tag{86}$$

where  $\Gamma_{J_l} = \sum_j \|\tilde{\alpha}_j\|$ ,  $\kappa(r) = \frac{1}{4\pi r^2} |1 - 4\pi q(r)|$  and  $\kappa_\sigma(r) = \frac{1}{\sigma^2} \kappa(\frac{r}{\sigma})$ ,  $r_j$  is the distance between source element and target elements,  $r_m$  is the distance that maximizes the kernel  $\kappa_\sigma(r)$ .

## Appendix B. Extra results

Figs. 18, 19, 20, and 21 show the  $L_2$  relative error norm of the velocity and vortex stretching vectors versus  $p$  using a second order algebraic core function and a second order Gaussian core function respectively. Figs. 22, 23, 24, and 25 show the  $L_2$  relative error norm of the velocity and vortex stretching vectors versus  $d_\sigma$  for a fixed order of expansion  $p = 5$  using a second order algebraic core function and a second order Gaussian core function respectively.

Fig. 26 shows the measured simulation time versus the number of elements for different  $n_F$  using an order of expansion  $p = 8$ .

## References

- [1] J.E. Barnes, P. Hut, A hierarchical  $o(n \log n)$  force calculations algorithm, *Nature* 324 (1986) 446.
- [2] R. Capuzzo-Dolcetta, P. Micocchi, A comparison between the fast multipole algorithm and the tree-code to evaluate gravitational forces in 3-d, *J. Comput. Phys.* 143 (1) (1998) 29–48.
- [3] H. Cheng, L. Greengard, V. Rokhlin, A fast adaptive multipole algorithm in three dimensions, *J. Comput. Phys.* 155 (2) (1999) 468–498.
- [4] W. Dehnen, A very fast and momentum-conserving tree code, *Astrophys. J.* 536 (2000) L39–L42.
- [5] W. Dehnen, A hierarchical  $o(n)$  force calculation algorithm, *arXiv preprint arXiv:astro-ph/0202512*, 2002.
- [6] L. Greengard, V. Rokhlin, A fast algorithm for particle simulations, *J. Comput. Phys.* 73 (1987) 325.
- [7] S. Salloum, I. Lakkis, Proper evaluation of spherical harmonics-based expressions for the velocity and vortex stretching vectors in three-dimensional grid-free vortex methods, *J. Comput. Phys.* (2020) 109603.
- [8] M.S. Warren, J.K. Salmon, A Portable Parallel Particle Program, *Comput. Phys. Commun.* 87 (1) (1995) 266–290, *Particle Simulation Methods*.
- [9] G. Winckelmans, A. Leonard, Contributions to vortex particle methods for the computation of three-dimensional incompressible unsteady flows, *J. Comput. Phys.* 109 (2) (1993) 247–273.
- [10] G. Winckelmans, U.C.D. Louvain, J.K. Salmon, A. Leonard, B. Jodoin, Application of Fast Parallel and Sequential Tree Codes to Computing Three-Dimensional Flows with the Vortex Element and Boundary Element Methods, 1996.
- [11] R. Yokota, An FMM based on dual tree traversal for many-core architectures, *CoRR*, arXiv:1209.3516 [abs], 2012.
- [12] R. Yokota, L. Barba, Parameter Tuning of a Hybrid Treecode-Fmm on Gpus, 2011.
- [13] R. Yokota, L. Barba, Hierarchical n-body simulations with autotuning for heterogeneous systems, *Comput. Sci. Eng.* 14 (3) (May 2012) 30–39.
- [14] R. Yokota, L.A. Barba, Fast n-body simulations on gpus, *CoRR*, arXiv:1108.5815 [abs], 2011.